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Image processing apparatus and image processing method

Abstract

The present disclosure relates to an image processing apparatus and an image processing method that are capable of improving coding efficiency at the time when a prediction using a correlation within a screen is performed. A predicted vector generation section sets, at a time of encoding of a current motion vector of a current block for a prediction using a correlation within a screen, a candidate block to not available and generates a predicted vector of the current motion vector by using a reference motion vector, the reference motion vector being referred to when the predicted vector of the current motion vector is generated. A difference vector generation section generates a difference vector between the current motion vector and the predicted vector generated by the predicted vector generation section. The present disclosure can be applied to, for example, an encoding apparatus.

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Background/Summary

CROSS REFERENCE TO PRIOR APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/217,513 (filed on Mar. 30, 2021), which is a continuation of U.S. patent application Ser. No. 16/861,377 (filed on Apr. 29, 2020), which is a continuation of U.S. patent application Ser. No. 15/516,992 (filed on Apr. 5, 2017 and issued as U.S. Pat. No. 10,735,762 on Aug. 4, 2020), which is a National Stage Patent Application of PCT International Patent Application No. PCT/JP2015/084763 (filed on Dec. 11, 2015) under 35 U.S.C. § 371, which claims priority to Japanese Patent Application No. 2014-265786 (filed on Dec. 26, 2014), which are all hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to an image processing apparatus and an image processing method, and more particularly to, an image processing apparatus and an image processing method that are capable of improving coding efficiency at the time when a prediction using a correlation within a screen is performed.

BACKGROUND ART

(2) There have been recently diffused apparatuses that handle image information as digital and, at that time, for the purpose of highly-efficient information transmission and accumulation, employ a coding system for compression by orthogonal transformation and motion compensation such as discrete cosine transform with use of redundancy peculiar to the image information, to perform compression coding on images. Examples of this coding system include MPEG (Moving Picture Experts Group), H.264, and MPEG-4 Part 10 (Advanced Video Coding, hereinafter described as AVC).

(3) Now, for the purpose of higher improvement in coding efficiency than H.264/AVC, standardization of a coding system called HEVC (High Efficiency Video Coding) is being advanced by JCTVC (Joint Collaboration Team-Video Coding), which is a joint standardization organization formed by ITU-T (International Telecommunication Union Telecommunication Standardization Sector) and ISO/IEC (International Organization for Standardization/International Electrotechnical Commission).

(4) Further, in HEVC, range extension (HEVC Range Extension) has been studied so as to support high-end formats, such as images in color difference signal formats like 4:2:2 and 4:4:4, profiles for screen content (see, for example, Non-Patent Literature 1).

(5) Incidentally, Intra Block Copy (Intra BC) is a coding tool of performing a prediction by using a correlation within a screen and performing motion compensation within the screen. Intra BC is known as a tool that contributes to improvement in coding efficiency for artificial images such as computer screens and CG images.

(6) However, the technique of Intra BC is not employed in the above-mentioned HEVC Range Extension extension and is continued to be studied for standardization of Screen Content Coding (SCC) extension (see, for example, Non-Patent Literature 2).

(7) As of December 2014, in the standardization of the HEVC SCC extension, commonalizing Intra BC and inter coding is in discussion. In the case where Intra BC and inter coding are commonalized, it is necessary to communalize selection of a predicted vector (MV Predictor), transmission of an index (RefIdx) of a reference picture, transmission of a difference vector (MVD) between a current motion vector and a predicted vector, and the like.

CITATION LIST

Non-Patent Literature

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DISCLOSURE OF INVENTION

Technical Problem

(9) However, in the case where a predicted vector generated with reference to a motion vector used in inter coding is used at the time of encoding of a motion vector used in Intra BC, a difference vector becomes large, and coding efficiency is reduced.

(10) The present disclosure has been made in view of the circumstances as described above and is capable of improving coding efficiency at the time when a prediction using a correlation within a screen is performed.

Solution to Problem

(11) An image processing apparatus according to a first aspect of the present disclosure is an image processing apparatus including: a predicted vector generation section that sets, at a time of encoding of a current motion vector of a current block for a prediction using a correlation within a screen, in a case where a type of a reference picture of the current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other, the candidate block to not available and generates a predicted vector of the current motion vector by using the reference motion vector, the reference motion vector being referred to when the predicted vector of the current motion vector is generated; and a difference vector generation section that generates a difference vector between the current motion vector and the predicted vector generated by the predicted vector generation section.

(12) An image processing method according to the first aspect of the present disclosure corresponds to the image processing apparatus according to the first aspect of the present disclosure.

(13) In the first aspect of the present disclosure, at the time of encoding of a current motion vector of a current block for a prediction using a correlation within a screen, in a case where a type of a reference picture of the current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other, the reference motion vector being referred to when a predicted vector of the current motion vector is generated, the candidate block is set to not available, the predicted vector of the current motion vector is generated by using the reference motion vector, and a difference vector between the current motion vector and the predicted vector is generated.

(14) An image processing apparatus according to a second aspect of the present disclosure is an image processing apparatus including: a predicted vector generation section that sets, at a time of decoding of a current motion vector of a current block for a prediction using a correlation within a screen, in a case where a type of a reference picture of the current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other, the candidate block to not available and generates a predicted vector of the current motion vector by using the reference motion vector, the reference motion vector being referred to when the predicted vector of the current motion vector is generated; and a motion vector generation section that adds a difference vector between the current motion vector and the predicted vector to the predicted vector generated by the predicted vector generation section and generates the current motion vector.

(15) An image processing method according to the second aspect of the present disclosure corresponds to the image processing apparatus according to the second aspect of the present disclosure.

(16) In the second aspect of the present disclosure, at the time of decoding of a current motion vector of a current block for a prediction using a correlation within a screen, in a case where a type of a reference picture of the current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other, the reference motion vector being referred to when a predicted vector of the current motion vector is generated, the candidate block is set to not available, the predicted vector of the current motion vector is generated by using the reference motion vector, a difference vector between the current motion vector and the predicted vector is added to the predicted vector, and the current motion vector is generated.

(17) It should be noted that the image processing apparatuses according to the first and second aspects can be achieved by causing a computer to execute a program.

(18) Further, the program executed by the computer in order to achieve the image processing apparatuses according to the first and second aspects can be provided by transmission via a transmission medium or recording on a recording medium.

(19) The image processing apparatuses according to the first and second aspects may be independent apparatuses or may be internal blocks that configure one apparatus.

Advantageous Effects of Invention

(20) According to the first aspect of the present disclosure, an image can be encoded. Further, according to the first aspect of the present disclosure, it is possible to improve coding efficiency at the time when a prediction using a correlation within a screen is performed.

(21) According to the second aspect of the present disclosure, an encoded stream can be decoded. Further, according to the second aspect of the present disclosure, it is possible to decode an encoded stream having improved coding efficiency at the time when a prediction using a correlation within a screen is performed.

(22) It should be noted that the effects described herein are not necessarily limited and any one of the effects described herein may be produced.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram for describing Intra BC.

(2) FIG. 2 is a diagram for describing a difference between Intra BC and inter coding.

(3) FIG. 3 is a diagram for describing a type of a reference picture in HEVC.

(4) FIG. 4 is a diagram showing an example of setting of a candidate block in HEVC.

(5) FIG. 5 is a diagram showing an example of setting of a candidate block in HEVC.

(6) FIG. 6 is a diagram for describing a candidate of a reference motion vector.

(7) FIG. 7 is a diagram showing an example of setting of a candidate block in a first embodiment.

(8) FIG. 8 is a diagram showing an example of setting of a candidate block in the first embodiment.

(9) FIG. 9 is a diagram showing an example of a candidate of a reference picture in HEVC.

(10) FIG. 10 is a diagram for describing reference picture list creation processing in HEVC.

(11) FIG. 11 is a diagram showing an example of a candidate of a reference picture in the first embodiment.

(12) FIG. 12 is a diagram for describing reference picture list creation processing in the first embodiment.

(13) FIG. 13 is a block diagram showing a configuration example of a first embodiment of an encoding apparatus to which the present disclosure is applied.

(14) FIG. 14 is a block diagram showing a configuration example of a coding section of FIG. 13.

(15) FIG. 15 is a flowchart for describing stream generation processing of the encoding apparatus of FIG. 13.

(16) FIG. 16 is a flowchart for describing details of encoding processing of FIG. 15.

(17) FIG. 17 is a flowchart for describing details of the encoding processing of FIG. 15.

(18) FIG. 18 is a flowchart for describing details of reference picture creation processing of FIG. 16.

(19) FIG. 19 is a flowchart for describing details of predicted vector list generation processing of FIG. 16.

(20) FIG. 20 is a block diagram showing a configuration example of a first embodiment of a decoding apparatus to which the present disclosure is applied.

(21) FIG. 21 is a block diagram showing a configuration example of a decoding section of FIG. 20.

(22) FIG. 22 is a flowchart for describing image generation processing of the decoding apparatus of FIG. 20.

(23) FIG. 23 is a flowchart for describing details of decoding processing of FIG. 22.

(24) FIG. 24 is a diagram for describing a predicted vector list generation method in a second embodiment.

(25) FIG. 25 is a diagram for describing the predicted vector list generation method in the second embodiment.

(26) FIG. **26** is a diagram for describing the predicted vector list generation method in the second embodiment.

(27) FIG. **27** is a diagram showing an example of setting of a candidate block in the second embodiment.

(28) FIG. **28** is a flowchart for describing details of reference picture creation processing in the second embodiment.

(29) FIG. **29** is a flowchart for describing details of predicted vector list generation processing in the second embodiment.

(30) FIG. **30** is a diagram for describing reference picture list creation processing in a third embodiment.

(31) FIG. **31** is a flowchart for describing details of the reference picture list creation processing in the third embodiment.

(32) FIG. **32** is a block diagram showing a configuration example of computer hardware.

(33) FIG. **33** is a diagram showing a schematic configuration example of a television apparatus to which the present disclosure is applied.

(34) FIG. **34** is a diagram showing a schematic configuration example of a mobile phone to which the present disclosure is applied.

(35) FIG. **35** is a diagram showing a schematic configuration example of a recording/reproduction apparatus to which the present disclosure is applied.

(36) FIG. **36** is a diagram showing a schematic configuration example of an imaging apparatus to which the present disclosure is applied.

(37) FIG. **37** shows a schematic configuration example of a video set to which the present disclosure is applied.

(38) FIG. **38** shows a schematic configuration example of a video processor to which the present disclosure is applied.

(39) FIG. **39** shows another schematic configuration example of the video processor to which the present disclosure is applied.

MODE(S) FOR CARRYING OUT THE INVENTION

(40) Hereinafter, the premise of the present disclosure and modes for carrying out the present disclosure (hereinafter, called embodiments) will be described. It should be noted that description will be given in the following order. 0. Premise of the Present Disclosure (FIGS. **1** and **2**) 1. First Embodiment: Encoding Apparatus and Decoding Apparatus (FIGS. **3** to **23**) 2. Second Embodiment: Encoding Apparatus and Decoding Apparatus (FIGS. **24** to **29**) 3. Third Embodiment: Encoding Apparatus and Decoding Apparatus (FIGS. **30** and **31**) 4. Fourth Embodiment: Computer (FIG. **32**) 5. Fifth Embodiment: Television Apparatus (FIG. **33**) 6. Sixth Embodiment: Mobile Phone (FIG. **34**) 7. Seventh Embodiment: Recording/reproduction apparatus (FIG. **35**) 8. Eighth Embodiment: Imaging Apparatus (FIG. **36**) 9. Ninth Embodiment: Video Set (FIGS. **37** to **39**)

Premise of the Present Disclosure

(41) (Description on Intra BC)

(42) FIG. **1** is a diagram for describing Intra Block Copy (Intra BC).

(43) As shown in FIG. **1**, in Intra BC, a reference picture used for predicting a PU (Prediction Unit) **11** within a picture **10** is the picture **10**. Therefore, a block **12**, which is included within the same picture **10** as the PU **11** and has a high correlation with the PU **11**, is set as a reference block of the PU **11**, and a vector **13** between the PU **11** and the PU **12** is detected as a motion vector.

(44) (Description on Difference Between Intra BC and Inter Coding)

(45) FIG. **2** is a diagram for describing a difference between Intra BC and inter coding.

(46) As shown in FIG. **2**, the precision of a motion vector in Intra BC is integer-pixel precision, and the precision of a motion vector in inter coding is $\frac{1}{4}$ pixel precision. Further, in Intra BC, since an encoded block within the same screen is set as a reference block, at least one of a length in a

longitudinal direction and a length in a lateral direction of the motion vector is longer. In inter coding, however, since an encoded block within a different screen is set as a reference block, the length of the motion vector depends on a motion between the screens.

(47) Further, in Intra BC, a picture before subjected to filter processing is used as a reference picture, but in inter coding, a picture after subjected to filter processing is used as a reference picture. In general, the picture before subjected to filter processing is stored in a cache, and the picture after subjected to filter processing is stored in a DRAM (Dynamic Random Access Memory).

(48) As described above, the precision and the length of the motion vector are different between Intra BC and inter coding. Therefore, if Intra BC and inter coding are communalized and a predicted vector generated with reference to the motion vector of inter coding is used at the time of encoding of the motion vector of Intra BC, a difference vector becomes large and coding efficiency is reduced. It should be noted that the motion vector referred to when the predicted vector is generated is hereinafter called a reference motion vector.

(49) Further, a destination to save the reference picture is different between Intra BC and inter coding. Therefore, in the case where Intra BC and inter coding are commonalized, the cache and the DRAM are mixed as the destinations to save the reference picture. However, since a decoding apparatus does not recognize its destination to save the reference picture, the decoding apparatus uselessly accesses the DRAM. As a result, the access bandwidth of the DRAM is increased.

First Embodiment

General Outline of First Embodiment

(50) First, the general outline of a first embodiment will be described with reference to FIGS. 3 to 12.

(51) FIG. 3 is a diagram for describing a type of a reference picture in HEVC (High Efficiency Video Coding).

(52) As shown in FIG. 3, in HEVC, a type of a reference picture that has a small difference in POC (Picture Order Count) from a picture to be currently processed (hereinafter, called current picture) is set as STRP (Short-Term Reference Picture). Further, a type of a reference picture that has a large difference in POC from the current picture (Current Pic) is set as LTRP (Long-Term Reference Picture).

(53) As shown in FIGS. 4 and 5, in the case where a reference picture of a PU to be currently processed (hereinafter, called current block) within the current picture and a reference picture of a PU (hereinafter, candidate block) corresponding to a candidate of a reference motion vector are different in type from each other, the candidate block is set to not available.

(54) For example, as shown in FIG. 3, in the case where the type of a reference picture of a current block 21 is STRP, and the type of a reference picture of a candidate block 22 corresponding to a candidate (PMV) of a reference motion vector of a current motion vector (target MV) is LTRP, the candidate block 22 is set to not available. Thus, the motion vector of the candidate block 22 is excluded from the candidates of the reference motion vector and is not used for generation of the predicted vector.

(55) It should be noted that candidate blocks are PUs that are adjacent to the current block on the left, the upper left, the lower left, the top, and the upper right at the same time of day, and are PUs that are located at the same position as the current block and adjacent thereto on the lower right at a different time of day. Hereinafter, in the case where a candidate block at the same time of day as the current block is especially distinguished from the other candidate blocks, the candidate block is called a spatial-direction candidate block, and in the case where a candidate block at a time of day different from that of the current block is especially distinguished from the other candidate blocks, the candidate block is called a temporal-direction candidate block.

(56) Further, as shown in FIGS. 4 and 5, in the case where the type of the reference picture of the current block and the type of the reference picture of the candidate block are identical to each other,

the candidate block is set to available. Thus, the motion vector of the candidate block is set to a candidate of the reference motion vector. In the case where the motion vector of the candidate block when both the type of the reference picture of the current block and the type of the reference picture of the candidate block are STRP is selected as a reference motion vector, the predicted vector is generated by scaling of the reference motion vector on the basis of a difference in POC between the reference picture of the current block and the reference picture of the candidate block. (57) Meanwhile, in the case where the motion vector of the candidate block when both the type of the reference picture of the current block and the type of the reference picture of the candidate block are LTRP is selected as a reference motion vector, the reference motion vector is set to a predicted vector without change (not scaled).

(58) It should be noted that Intra BC and inter coding are not commonalized in the present HEVC, and thus the type of a reference picture other than LTRP and STRP is not assumed as shown in FIG. 5. Therefore, in the case where the type of a reference picture is “Intra BC”, which is the type of a reference picture used in Intra BC, setting of a candidate block is undetermined.

(59) In the case where Intra BC and inter coding are simply communalized, the type of a reference picture used in Intra BC is different from an actual one and is set to STRP. Therefore, as shown in FIG. 6, a motion vector (PMV) of a candidate block 32, the actual type of a reference picture of which is STRP, is included in candidates of a reference motion vector of a current motion vector (target MV) used in Intra BC for a current block 31.

(60) As shown in FIG. 2, however, the precision and the length of the motion vector are different between Intra BC and inter coding. Therefore, if the motion vector of the candidate block 32 is selected as the reference motion vector of the current motion vector of the current block 31, a difference vector between the current motion vector and a predicted vector becomes large. As a result, coding efficiency is reduced.

(61) In this regard, in the first embodiment, the type of the reference picture used in Intra BC, i.e., the type of the reference picture the actual type of which is “Intra BC”, is set to LTRP. Specifically, a current picture is newly defined as a candidate of the reference picture, (If `intra_block_copy_enabled_flag` is equal to 1, the following applies. `RefPicCurr=CurrPic`). The current picture newly defined is then added as a candidate of the reference picture, the type of which is set to LTRP (All reference pictures that are included in `RefPicCurr`, `RefPicSetLtCurr` and `RefPicSetLtFoll` are marked as “used for long-term reference”).

(62) Thus, the current picture is added to a reference picture list, which is a list of candidates of the reference picture, and is marked as “used for long-term reference”. As a result, setting of a candidate block with respect to the current block the actual type of the reference picture of which is “Intra BC” is performed similarly to setting of a candidate block with respect to the current block the actual type of the reference picture of which is LTRP.

(63) Specifically, as shown in FIGS. 7 and 8, in the case where any one of the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block is “Intra BC” and the other one is STRP, the candidate block is set to not available. Thus, the motion vector of the candidate block is excluded from the candidates of the reference motion vector and is not used for generation of the predicted vector.

(64) Further, as shown in FIGS. 7 and 8, in the case where both the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block are “Intra BC” or LTRP, the candidate block is set to available. Thus, the motion vector of the candidate block is set to a candidate of the reference motion vector.

(65) As described above, in the case where any one of the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block is “Intra BC” and the other one is STRP, the motion vector of the candidate block is not used for generation of the predicted vector and coding efficiency is improved.

(66) FIG. 9 is a diagram showing an example of a candidate of the reference picture in HEVC.

(67) As shown in FIG. 9, in HEVC, pictures (Short-term before Curr) that are previous to a current picture (C) in display order and have a small difference in POC from the current picture are candidates of the reference picture. Further, pictures (Short-term after Curr) that are subsequent to the current picture (C) in display order and have a small difference in POC from the current picture are candidates of the reference picture.

(68) Moreover, pictures (Long-term) that are previous to the current picture (C) in display order and have a large difference in POC from the current picture are candidates of the reference picture. Further, pictures (Inter-layer) that are the same as the current picture (C) in display order but different in perspective, bit count, spatial resolution, S/N, frame rate, color difference signal format, and the like are candidates of the reference picture.

(69) FIG. 10 is a diagram for describing reference picture list creation processing for registering the candidates of the reference picture of FIG. 9.

(70) As shown in FIG. 10, in the reference picture list creation processing, first, an individual list in which information for identifying a candidate picture of the reference picture (hereinafter, called picture identifying information) is registered is created for each kind of candidates of the reference picture. In other words, a RefPicSetStCurrBefore list is created, in which the picture identifying information of each of the pictures (Short-term before Curr) is registered in ascending order of the difference in POC from the current picture.

(71) Similarly, a RefPicSetStCurrAfter list of the pictures (Short-term after Curr) and a RefPicSetLtCurr list of the pictures (Long-term) are created. Further, a RefPicSetIvCurr list is created, in which the picture identifying information of each of the pictures (Inter-layer) is registered in predetermined order.

(72) For each piece of the picture identifying information registered in the RefPicSetStCurrBefore list, the RefPicSetStCurrAfter list, and the RefPicSetLtCurr list, used_by_curr can be set. The used_by_curr indicates whether a picture identified by that picture identifying information is used as a candidate of the reference picture of the current picture. The used_by_curr is 1 when indicating that the picture is used as a candidate of the reference picture of the current picture, and is 0 when indicating that the picture is not used.

(73) In the example of FIG. 10, the used_by_curr of the head and second pieces of the picture identifying information of the RefPicSetStCurrBefore list and the RefPicSetStCurrAfter list and the head piece of the picture identifying information of the RefPicSetLtCurr list are set to 1.

(74) Next, on the basis of the individual lists, two temporal lists, i.e., a RefPicListTemp0 list that preferentially registers the candidates of the reference picture that are previous to the current picture in display order and a RefPicListTemp1 list that preferentially registers the candidates of the reference picture that are subsequent to the current picture in display order, are created.

(75) Specifically, the RefPicListTemp0 list is created, which registers in order the pieces of the picture identifying information of the RefPicSetStCurrBefore list, for which the used_by_curr is set to 1, the pieces of the picture identifying information registered in the RefPicSetIvCurr list, the pieces of the picture identifying information of the RefPicSetStCurrAfter list, for which the used_by_curr is set to 1, and the piece of the picture identifying information of the RefPicSetLtCurr list, for which the used_by_curr is set to 1.

(76) Further, the RefPicListTemp1 list is created, which registers in order the pieces of the picture identifying information of the RefPicSetStCurrAfter list, for which the used_by_curr is set to 1, the pieces of the picture identifying information of the RefPicSetStCurrBefore list, for which the used_by_curr is set to 1, the piece of the picture identifying information of the RefPicSetLtCurr list, for which the used_by_curr is set to 1, and the pieces of the picture identifying information registered in the RefPicSetIvCurr list.

(77) Next, the order of the pieces of the picture identifying information registered in the RefPicListTemp0 list and RefPicListTemp1 list is changed as needed (Reference reordering).

(78) Lastly, a reference picture list RefPicList0 is created, in which a predetermined number (five

in the example of FIG. 10) of pieces of the picture identifying information from the head, which are registered in the RefPicListTemp0 list, are registered in order from the head.

(79) Further, a reference picture list RefPicList1 is created, in which a predetermined number (four in the example of FIG. 10) of pieces of the picture identifying information from the head, which are registered in the RefPicListTemp1 list, are registered in order from the head.

(80) It should be noted that a value obtained by subtracting 1 from the number of pieces of the picture identifying information registered in the reference picture list RefPicList0 (RefPicList1) is disposed at a slice header or the like as num_ref_idx_10_active_minus1 (num_ref_idx_11_active_minus1).

(81) In the case where Intra BC and inter coding are commonalized, as shown in FIG. 11, it is necessary to add a current picture (CurrPic), which is the reference picture in Intra BC, as a candidate of the reference picture.

(82) In the first embodiment, as described above, since the current picture is defined as a candidate of the reference picture, a current picture list (RefPicCurr) in which the picture identifying information of the current picture is registered can be created in the reference picture list creation processing. As a result, the picture identifying information of the current picture is registered in the reference picture list.

(83) In the case where Intra BC and inter coding are simply communalized, however, it is unknown in which position of the reference picture list the picture identifying information of the current picture is registered.

(84) Meanwhile, as shown in FIG. 2, the destination to save the reference picture is different between Intra BC and inter coding. Therefore, in order not to uselessly access the DRAM, when a decoding apparatus acquires an index of the picture identifying information of the reference picture within the reference picture list from an encoding apparatus, the decoding apparatus needs to recognize that the reference picture is a reference picture in any of Intra BC and inter coding. In other words, the decoding apparatus needs to recognize that the reference picture is any of the current picture and a picture different from the current picture.

(85) In this regard, in the first embodiment, as shown in FIG. 12, the picture identifying information of the current picture, which is registered in the current picture list, is registered in the heads of the two reference picture lists, the RefPicList0 list and the RefPicList1 list.

(86) Thus, in the case where the decoding apparatus acquires the index in the head of the reference picture list from the encoding apparatus, the decoding apparatus can recognize that the reference picture is the current picture, i.e., the reference picture in Intra BC. As a result, the decoding apparatus can read the current picture, as a reference picture, from the cache without accessing the DRAM.

(87) It should be noted that the picture identifying information of the current picture may be registered in the RefPicList0 list only. Further, the position at which the picture identifying information of the current picture is registered can be set to an arbitrary predetermined position determined in advance, such as the tail end, in addition to the head.

(88) (Configuration Example of First Embodiment of Encoding Apparatus)

(89) FIG. 13 is a block diagram showing a configuration example of a first embodiment of an encoding apparatus as an image processing apparatus to which the present disclosure is applied.

(90) An encoding apparatus 50 of FIG. 13 includes a setting section 51, an encoding section 52, and a transmission section 53 and encodes a picture by a system according to HEVC.

(91) A unit of the encoding is a Coding UNIT (CU) having a recursive hierarchical structure. Specifically, the CU is set by dividing a picture into CTUs (Coding Tree Units) having a fixed size and dividing the CTU into two by arbitrary times in the horizontal direction and vertical direction. The maximum size of the CU is an LCU (Largest Coding Unit), and the minimum size thereof is an SCU (Smallest Coding Unit). Further, the CU is divided into PUs or TUs (transform units).

(92) The setting section 51 of the encoding apparatus 50 sets a parameter set such as an SPS

(Sequence Parameter Set), a PPS (Picture Parameter Set), VUI (Video Usability Information), and SEI (Supplemental Enhancement Information). The SPS includes information specifying the size of the LCU and the SCU, reference picture list information for creating a reference picture list, and the like.

(93) The reference picture list information is constituted of, for example, information specifying the picture identifying information registered in the individual lists and the current picture list, `used_by_curr`, and the like. The setting section **51** supplies the set parameter set to the encoding section **52**.

(94) Pictures in units of frame are input to the encoding section **52**. The encoding section **52** encodes the input pictures by a system according to HEVC. At that time, the parameter set supplied from the setting section **51** is used as needed. The encoding section **52** generates an encoded stream from encoded data obtained as a result of encoding and the parameter set supplied from the setting section **51** and supplies the encoded stream to the transmission section **53**.

(95) The transmission section **53** transmits the encoded stream, which is supplied from the encoding section **52**, to a decoding apparatus that will be described later.

(96) (Configuration Example of Encoding Section)

(97) FIG. **14** is a block diagram showing a configuration example of the encoding section **52** of FIG. **13**.

(98) The encoding section **52** of FIG. **14** includes an A/D conversion section **71**, a screen sorting buffer **72**, a computing section **73**, an orthogonal transformation section **74**, a quantization section **75**, a lossless encoding section **76**, an accumulation buffer **77**, a generation section **78**, an inverse quantization section **79**, an inverse orthogonal transformation section **80**, and an adder section **81**. Further, the encoding section **52** includes a filter **82**, a frame memory **85**, a switch **86**, an intra prediction section **87**, a list creation section **88**, a motion prediction/compensation section **89**, a predicted vector generation section **90**, a difference vector generation section **91**, a predicted image selection section **92**, and a rate control section **93**.

(99) The A/D conversion section **71** of the encoding section **52** performs A/D conversion on pictures in unit of frame, which are input as encoding targets. The A/D conversion section **71** outputs the pictures, as digital signals after conversion, to the screen sorting buffer **72** to be stored therein.

(100) The screen sorting buffer **72** sorts the pictures in unit of frame, which are stored in display order, in encoding order in accordance with the GOP (Group of Picture) structure. The screen sorting buffer **72** outputs each of the sorted pictures, as a current picture, to the computing section **73**, the intra prediction section **87**, and the motion prediction/compensation section **89**.

(101) The computing section **73** performs encoding by subtracting a predicted image, which is supplied from the predicted image selection section **92**, from the current picture supplied from the screen sorting buffer **72**. The computing section **73** outputs the resultant picture, as residual information, to the orthogonal transformation section **74**. It should be noted that in the case where the predicted image is not supplied from the predicted image selection section **92**, the computing section **73** outputs the current picture read from the screen sorting buffer **72**, as residual information, to the orthogonal transformation section **74** without change.

(102) The orthogonal transformation section **74** performs orthogonal transformation on the residual information from the computing section **73** in units of TU. The orthogonal transformation section **74** supplies an orthogonal transformation coefficient obtained as a result of the orthogonal transformation to the quantization section **75**.

(103) The quantization section **75** quantizes the orthogonal transformation coefficient supplied from the orthogonal transformation section **74**. The quantization section **75** supplies the quantized orthogonal transformation coefficient to the lossless encoding section **76**.

(104) The lossless encoding section **76** acquires intra prediction mode information, which indicates an optimal intra prediction mode, from the intra prediction section **87**. Further, the lossless

encoding section **76** acquires inter prediction mode information, which indicates an optimal inter prediction mode, an index of the reference picture, and the like from the motion prediction/compensation section **89**, and acquires motion vector information from the difference vector generation section **91**. Moreover, the lossless encoding section **76** acquires offset filter information on adaptive offset filter processing from the filter **82**.

(105) The lossless encoding section **76** performs lossless encoding, such as variable-length encoding (for example, CAVLC (Context-Adaptive Variable Length Coding) etc.) and arithmetic encoding (for example, CABAC (Context-Adaptive Binary Arithmetic Coding) etc.), on the quantized orthogonal transformation coefficient supplied from the quantization section **75**.

(106) Further, the lossless encoding section **76** performs lossless encoding on the intra prediction mode information or the inter prediction mode information, the motion vector information, the index of the reference picture, the offset filter information, and the like, as encoding information on encoding. The lossless encoding section **76** disposes the lossless-encoded encoding information, a predetermined number (num_ref_idx_lo_minus1, num_ref_idx_lo_minus1) obtained by subtracting 1 from a registration number of the reference picture list, and the like in the slice header or the like.

(107) The lossless encoding section **76** adds the slice header to the orthogonal transformation coefficient supplied from the quantization section **75** and then supplies the resultant data, as encoded data, to the accumulation buffer **77**.

(108) The accumulation buffer **77** temporarily stores the encoded data supplied from the lossless encoding section **76**. Further, the accumulation buffer **77** supplies the stored encoded data to the generation section **78**.

(109) The generation section **78** generates an encoded stream from the parameter set supplied from the setting section **51** of FIG. **13** and the encoded data supplied from the accumulation buffer **77** and supplies the encoded stream to the transmission section **53** of FIG. **13**.

(110) Further, the quantized orthogonal transformation coefficient output from the quantization section **75** is also input to the inverse quantization section **79**. The inverse quantization section **79** inversely quantizes the orthogonal transformation coefficient, which is quantized by the quantization section **75**, by a method corresponding to a quantization method of the quantization section **75**. The inverse quantization section **79** supplies the orthogonal transformation coefficient obtained as a result of inverse quantization to the inverse orthogonal transformation section **80**.

(111) The inverse orthogonal transformation section **80** performs inverse orthogonal transformation on the orthogonal transformation coefficient, which is supplied from the inverse quantization section **79**, in units of TU by a method corresponding to an orthogonal transformation method of the orthogonal transformation section **74**. The inverse orthogonal transformation section **80** supplies the resultant residual information to the adder section **81**.

(112) The adder section **81** adds the residual information supplied from the inverse orthogonal transformation section **80** and the predicted image supplied from the predicted image selection section **92**, to locally decode the current picture. It should be noted that in the case where the predicted image is not supplied from the predicted image selection section **92**, the adder section **81** sets the residual information, which is supplied from the inverse orthogonal transformation section **80**, to a decoding result. The adder section **81** supplies the locally-decoded current picture to the frame memory **85**. Further, the adder section **81** supplies the current picture, all regions of which are decoded, as an encoded picture, to the filter **82**.

(113) The filter **82** performs filter processing on the encoded picture supplied from the adder section **81**. Specifically, the filter **82** performs deblocking filter processing and adaptive offset filter (SAO (Sample adaptive offset)) processing in sequence. The filter **82** supplies the encoded picture after subjected to the filter processing to the frame memory **85**. Further, the filter **82** supplies information indicating an offset and a kind of the adaptive offset filter processing performed, as offset filter information, to the lossless encoding section **76**.

(114) The frame memory **85** is constituted of, for example, a cache and a DRAM. The frame

memory **85** stores the current picture, which is supplied from the adder section **81**, in the cache and stores the encoded picture, which is supplied from the filter **82**, in the DRAM. Pixels adjacent to a current block in the current picture stored in the cache are set as peripheral pixels and supplied to the intra prediction section **87** via the switch **86**.

(115) Further, the current picture stored in the cache and the encoded picture stored in the DRAM are set as reference pictures and output to the motion prediction/compensation section **89** via the switch **86**.

(116) The intra prediction section **87** performs intra prediction processing on the current block in all candidate intra prediction modes by using the peripheral pixels read from the frame memory **85** via the switch **86**.

(117) Further, the intra prediction section **87** calculates a cost function value (details thereof will be described later) for all candidate intra prediction modes on the basis of the current picture read from the screen sorting buffer **72** and a predicted image generated as a result of the intra prediction processing. The intra prediction section **87** then determines an intra prediction mode, the cost function value of which is the smallest, to be the optimal intra prediction mode.

(118) The intra prediction section **87** supplies the predicted image, which is generated in the optimal intra prediction mode, and a corresponding cost function value to the predicted image selection section **92**. In the case where the intra prediction section **87** is notified of a selection of the predicted image generated in the optimal intra prediction mode by the predicted image selection section **92**, the intra prediction section **87** supplies the intra prediction mode information to the lossless encoding section **76**.

(119) It should be noted that the cost function value is also called an RD (Rate Distortion) cost and is calculated on the basis of, for example, a technique of a High Complexity mode or Low Complexity mode as determined in JM (Joint Model), which is reference software in the H.264/AVC system. It should be noted that the reference software in the H.264/AVC system is publicly available in <http://iphome.hhi.de/suehring/tml/index.htm>.

(120) Specifically, in the case where the High Complexity mode is employed as a technique of calculating the cost function value, all candidate prediction modes are temporarily subjected to processing to decoding, and a cost function value expressed by the following expression (1) is calculated for each prediction mode.

(121) [MathematicalExpression1]
$$\text{Cost}(\text{Mode}) = D + \lambda \cdot \text{Math. } R \quad (1)$$

(122) D represents a difference (distortion) between a raw image and a decoded image, R represents the amount of generated codes including a coefficient of orthogonal transformation as well, and λ represents a Lagrange multiplier given as a function of a quantization parameter QP.

(123) Meanwhile, in the case where the Low Complexity mode is employed as a technique of calculating the cost function value, generation of a predicted image and calculation of the amount of codes of encoding information are performed for all candidate prediction modes, and a cost function Cost (Mode) expressed by the following expression (2) is calculated for each prediction mode.

(124) [MathematicalExpression2]
$$\text{Cost}(\text{Mode}) = D + \text{QPtoQuant}(\text{QP}) \cdot \text{Math. Header_Bit} \quad (2)$$

(125) D represents a difference (distortion) between a raw image and a predicted image, Header_Bit represents the amount of codes of encoding information, and QPtoQuant represents a function given as a function of a quantization parameter QP.

(126) In the Low Complexity mode, it is only necessary to generate the predicted images for all the prediction modes and unnecessary to generate a decoded image. This leads to a smaller amount of computation.

(127) The list creation section **88** performs the above-mentioned reference picture list creation

processing, for example, in units of slice, on the basis of the reference picture list information included in the SPS set by the setting section **51** of FIG. **13**, the information disposed in the slice header, and the like.

(128) In other words, the list creation section **88** performs in sequence creation of the individual lists, creation of the temporal lists, rearrangement of the order of the pieces of the picture identifying information registered in the temporal lists, on the basis of the reference picture list information, the information disposed in the slice header, and the like, to create the reference picture list. The list creation section **88** then registers the picture identifying information of the current picture in the head of the reference picture list.

(129) Further, the list creation section **88** sets SRTP or LTRP for each piece of the picture identifying information registered in the reference picture list, as a type of a reference picture of a picture identified by that picture identifying information. For example, the list creation section **88** (setting section) sets LTRP (used for long-term reference), as a type of the current picture, for the picture identifying information of the current picture. The list creation section **88** holds the reference picture list and also supplies the reference picture list to the motion prediction/compensation section **89**.

(130) The motion prediction/compensation section **89** performs motion prediction/compensation processing on the current block in all candidate inter prediction modes. Specifically, on the basis of the reference picture list supplied from the list creation section **88**, the motion prediction/compensation section **89** reads a picture, which is identified by picture identifying information registered in that reference picture list, as a candidate of the reference picture, from the frame memory **85** via the switch **86**.

(131) On the basis of the current picture from the screen sorting buffer **72** and the current picture read as a candidate of the reference picture, the motion prediction/compensation section **89** detects in integer-pixel precision a motion vector of Intra BC of the current block in all candidate inter prediction modes. Further, on the basis of the current picture from the screen sorting buffer **72** and a picture other than the current picture read as a candidate of the reference picture, the motion prediction/compensation section **89** detects in a fraction-pixel precision a motion vector of inter coding of the current block in all candidate inter prediction modes.

(132) On the basis of the detected motion vectors, the motion prediction/compensation section **89** then performs compensation processing on the candidates of the reference picture and generates a predicted image. It should be noted that the inter prediction mode is a mode indicating the size of the current block and the like.

(133) Further, the motion prediction/compensation section **89** calculates a cost function value for all candidate inter prediction modes and the candidates of the reference picture on the basis of the current picture supplied from the screen sorting buffer **72** and the predicted image. The motion prediction/compensation section **89** determines an inter prediction mode, the cost function value of which is the smallest, to be the optimal inter prediction mode. The motion prediction/compensation section **89** then supplies the smallest cost function value and a corresponding predicted image to the predicted image selection section **92**.

(134) In the case where the motion prediction/compensation section **89** is notified of a selection of the predicted image generated in the optimal inter prediction mode by the predicted image selection section **92**, the motion prediction/compensation section **89** determines a motion vector of the current block corresponding to that predicted image to be a current motion vector and supplies the motion vector to the difference vector generation section **91**. Further, the motion prediction/compensation section **89** determines a candidate of the reference picture corresponding to that predicted image to be a reference picture, and supplies an index corresponding to a position, in the reference picture list, of picture identifying information of that reference picture to the predicted vector generation section **90** and the lossless encoding section **76**. Moreover, the motion prediction/compensation section **89** supplies the inter prediction mode information to the lossless

encoding section **76**.

(135) The predicted vector generation section **90** generates a predicted vector list of the current motion vector for each reference picture list in which the reference picture is registered, on the basis of the index supplied from the motion prediction/compensation section **89**.

(136) Specifically, on the basis of the index, the predicted vector generation section **90** reads a type of the reference picture corresponding to that index from the reference picture list held in the list creation section **88**, and holds the type. The predicted vector generation section **90** sets the spatial-direction candidate block to a candidate block to be processed, in predetermined order, and sets the temporal-direction candidate block to a candidate block to be processed, in predetermined order.

(137) The predicted vector generation section **90** compares the type of the reference picture of the candidate block to be processed, which is held therein, and the type of the reference picture of the current block read from the list creation section **88**. In the case where those types are matched with each other, the predicted vector generation section **90** sets the candidate block to be processed to be available. In the case where those types are not matched with each other, the predicted vector generation section **90** sets the candidate block to be processed to not available.

(138) In the case where the candidate block to be processed is set to available, the predicted vector generation section **90** generates a predicted vector of the current motion vector by using a candidate of the reference motion vector that is a motion vector of the candidate block to be processed. The predicted vector generation section **90** then generates a predicted vector list that registers a predetermined number of generated predicted vectors of the spatial-direction candidate blocks and temporal-direction candidate blocks. It should be noted that the number of generated predicted vectors does not reach a predetermined number, a zero vector or the like is registered in the predicted vector list. The predicted vector generation section **90** supplies the predicted vector list to the difference vector generation section **91**.

(139) The difference vector generation section **91** generates a difference vector between each of a predetermined number of predicted vectors, which are registered in the predicted vector list supplied from the predicted vector generation section **90**, and the current motion vector, to perform prediction encoding on the current motion vector. The difference vector generation section **91** supplies, the smallest value of the difference vector and an index corresponding to a position, in the predicted vector list, of a corresponding predicted vector, to the lossless encoding section **76**.

(140) On the basis of the cost function values supplied from the intra prediction section **87** and the motion prediction/compensation section **89**, the predicted image selection section **92** determines a mode having a smaller corresponding cost function value, in the optimal intra prediction mode and the optimal inter prediction mode, to be the optimal prediction mode. The predicted image selection section **92** then supplies the predicted image of the optimal prediction mode to the computing section **73** and the adder section **81**. Further, the predicted image selection section **92** notifies the intra prediction section **87** or the motion prediction/compensation section **89** of a selection of the predicted image of the optimal prediction mode.

(141) The rate control section **93** controls the rate of a quantization operation of the quantization section **75** on the basis of the encoded data accumulated in the accumulation buffer **77** such that an overflow or underflow does not occur.

(142) (Description on Processing of Encoding Apparatus)

(143) FIG. **15** is a flowchart for describing stream generation processing of the encoding apparatus **50** of FIG. **13**.

(144) In Step **S11** of FIG. **15**, the setting section **51** of the encoding apparatus **50** sets a parameter set. The setting section **51** supplies the set parameter set to the encoding section **52**.

(145) In Step **S12**, the encoding section **52** performs encoding processing of encoding pictures in units of frame, which are input from the outside, by a system according to HEVC. Details of the encoding processing will be described with reference to FIGS. **16** and **17** that will be described later.

(146) In Step S13, the generation section **78** (FIG. **14**) of the encoding section **52** generates an encoded stream from the parameter set supplied from the setting section **51** and accumulated encoded data and supplies the encoded stream to the transmission section **53**.

(147) In Step S14, the transmission section **53** transmits the encoded stream, which is supplied from the setting section **51**, to a decoding apparatus that will be described later, and then terminates the processing.

(148) FIGS. **16** and **17** are flowcharts for describing details of the encoding processing in Step S12 of FIG. **15**.

(149) In Step S31 of FIG. **16**, the A/D conversion section **71** of the encoding section **52** performs A/D conversion on pictures in units of frame, which are input as encoding targets, and outputs the pictures, as digital signal after conversion, to the screen sorting buffer **72** to be stored therein.

(150) In Step S32, the screen sorting buffer **72** sorts the pictures of frames, which are stored in display order, in encoding order in accordance with the GOP structure. The screen sorting buffer **72** supplies each of the pictures in units of frame after the sorting, as a current picture, to the computing section **73**, the intra prediction section **87**, and the motion prediction/compensation section **89**.

(151) In Step S33, the list creation section **88** performs reference picture list creation processing, for example, in units of slice, on the basis of reference picture list information included in an SPS, information disposed in the slice header, and the like. Details of the reference picture list creation processing will be described with reference to FIG. **18**.

(152) In Step S34, the intra prediction section **87** performs intra prediction processing on a current block in all candidate intra prediction modes by using peripheral pixels read from the frame memory **85** via the switch **86**. Further, the intra prediction section **87** calculates a cost function value for all candidate intra prediction modes on the basis of the current picture from the screen sorting buffer **72** and a predicted image generated as a result of the intra prediction processing. The intra prediction section **87** then determines an intra prediction mode, the cost function value of which is the smallest, to be the optimal intra prediction mode. The intra prediction section **87** supplies the predicted image, which is generated in the optimal intra prediction mode, and a corresponding cost function value to the predicted image selection section **92**.

(153) Further, the motion prediction/compensation section **89** performs motion prediction/compensation processing on the current block in all candidate inter prediction modes on the basis of a reference picture list supplied from the list creation section **88**. Further, the motion prediction/compensation section **89** calculates a cost function value for all candidate inter prediction modes and the reference picture on the basis of the current picture supplied from the screen sorting buffer **72** and a predicted image generated as a result of the motion prediction/compensation processing. The motion prediction/compensation section **89** determines an inter prediction mode, the cost function value of which is the smallest, to be the optimal inter prediction mode. The motion prediction/compensation section **89** then supplies the smallest cost function value and a corresponding predicted image to the predicted image selection section **92**.

(154) In Step S35, the predicted image selection section **92** determines a mode having the smallest cost function value, in the optimal intra prediction mode and the optimal inter prediction mode, to be the optimal prediction mode on the basis of the cost function values supplied from the intra prediction section **87** and the motion prediction/compensation section **89**. The predicted image selection section **92** then supplies the predicted image of the optimal prediction mode to the computing section **73** and the adder section **81**.

(155) In Step S36, the predicted image selection section **92** determines whether the optimal prediction mode is the optimal inter prediction mode or not. In the case where it is determined in Step S36 that the optimal prediction mode is the optimal inter prediction mode, the predicted image selection section **92** notifies the motion prediction/compensation section **89** of a selection of the predicted image generated in the optimal inter prediction mode.

(156) In accordance with this notification, the motion prediction/compensation section **89** determines a motion vector of the current block corresponding to the predicted image to be a current motion vector and supplies the current motion vector to the difference vector generation section **91**. Further, the motion prediction/compensation section **89** determines a candidate of the reference picture corresponding to the predicted image to be a reference picture and supplies an index corresponding to a position, in the reference picture list, of picture identifying information of that reference picture to the predicted vector generation section **90**.

(157) In Step **S37**, the predicted vector generation section **90** performs predicted vector list generation processing of generating a predicted vector list of the current motion vector, on the basis of the index supplied from the motion prediction/compensation section **89**. Details of the predicted vector list generation processing will be described later with reference to FIG. **19**.

(158) In Step **S38**, the difference vector generation section **91** generates a difference vector between each of a predetermined number of predicted vectors, which are registered in the predicted vector list supplied from the predicted vector generation section **90**, and the current motion vector, to perform prediction encoding on the current motion vector. The difference vector generation section **91** then generates the smallest value of the difference vector and an index corresponding to a position, in the predicted vector list, of a corresponding predicted vector, as motion vector information.

(159) In Step **S39**, the motion prediction/compensation section **89** supplies the inter prediction mode information and the index of the reference picture to the lossless encoding section **76**, and the difference vector generation section **91** supplies the motion vector information to the lossless encoding section **76**. The processing then proceeds to Step **S41**.

(160) Meanwhile, in the case where it is determined in Step **S36** that the optimal prediction mode is not the optimal inter prediction mode, that is, in the case where the optimal prediction mode is the optimal intra prediction mode, the predicted image selection section **92** notifies the intra prediction section **87** of a selection of the predicted image generated in the optimal intra prediction mode. In Step **S40**, the intra prediction section **87** supplies the intra prediction mode information to the lossless encoding section **76**, and advances the processing to Step **S41**.

(161) In Step **S41**, the computing section **73** performs encoding by subtracting the predicted image, which is supplied from the predicted image selection section **92**, from the current picture supplied from the screen sorting buffer **72**. The computing section **73** outputs the resultant picture, as residual information, to the orthogonal transformation section **74**.

(162) In Step **S42**, the orthogonal transformation section **74** performs orthogonal transformation on the residual information from the computing section **73** in units of TU, and supplies a resultant orthogonal transformation coefficient to the quantization section **75**.

(163) In Step **S43**, the quantization section **75** quantizes the orthogonal transformation coefficient supplied from the orthogonal transformation section **74** and supplies the quantized orthogonal transformation coefficient to the lossless encoding section **76** and the inverse quantization section **79**.

(164) In Step **S44** of FIG. **17**, the inverse quantization section **79** inversely quantizes the quantized coefficient supplied the quantization section **75** and supplies the resultant orthogonal transformation coefficient to the inverse orthogonal transformation section **80**.

(165) In Step **S45**, the inverse orthogonal transformation section **80** performs inverse orthogonal transformation on the orthogonal transformation coefficient supplied from the inverse quantization section **79** in units of TU and supplies the resultant residual information to the adder section **81**.

(166) In Step **S46**, the adder section **81** adds the residual information supplied from the inverse orthogonal transformation section **80** and the predicted image supplied from the predicted image selection section **92**, to locally decode the current picture. The adder section **81** supplies the locally-decoded current picture to the frame memory **85**. Further, the adder section **81** supplies the current picture, all regions of which are decoded, as an encoded picture, to the filter **82**.

(167) In Step S47, the filter **82** performs deblocking filter processing on the encoded picture supplied from the adder section **81**.

(168) In Step S48, the filter **82** performs adaptive offset filter processing on the encoded picture after subjected to the deblocking filter processing for each of the LCUs. The filter **82** supplies the resultant encoded picture to the frame memory **85**. Further, the filter **82** supplies offset filter information to the lossless encoding section **76** for each of the LCUs.

(169) In Step S49, the frame memory **85** stores the current picture, which is supplied from the adder section **81**, in the cache and stores the encoded picture, which is supplied from the filter **82**, in the DRAM. Pixels adjacent to the current block in the current picture stored in the cache are set as peripheral pixels and supplied to the intra prediction section **87** via the switch **86**. Further, the current picture stored in the cache and the encoded picture stored in the DRAM are set as reference pictures and output to the motion prediction/compensation section **89** via the switch **86**.

(170) In Step S50, the lossless encoding section **76** performs lossless encoding on the intra prediction mode information or the inter prediction mode information, the motion vector information, the index of the reference picture, and the offset filter information, as encoding information.

(171) In Step S51, the lossless encoding section **76** performs lossless encoding on the quantized orthogonal transformation coefficient supplied from the quantization section **75**. The lossless encoding section **76** then generates encoded data from the encoding information lossless-encoded by the processing of Step S50 and the lossless-encoded orthogonal transformation coefficient, and supplies the encoded data to the accumulation buffer **77**.

(172) In Step S52, the accumulation buffer **77** temporarily accumulates the encoded data supplied from the lossless encoding section **76**.

(173) In Step S53, the rate control section **93** controls the rate of a quantization operation of the quantization section **75** on the basis of the encoded data accumulated in the accumulation buffer **77** such that an overflow or underflow does not occur.

(174) In Step S54, the accumulation buffer **77** outputs the stored encoded data to the generation section **78**. The processing then returns to Step S12 of FIG. 15 and proceeds to Step S13.

(175) It should be noted that, for the purpose of simplifying description, the intra prediction processing and the motion prediction/compensation processing are constantly performed in the encoding processing of FIGS. 16 and 17, but actually, any one of the processing is performed depending on a picture type or the like in some cases.

(176) FIG. 18 is a flowchart for describing details of the reference picture list creation processing in Step S33 of FIG. 16.

(177) In Step S71 of FIG. 18, the list creation section **88** creates four individual lists, i.e., a RefPicSetStCurrBefore list, a RefPicSetStCurrAfter list, a RefPicSetLtCurr list, and a RefPicSetIvCurr list for each kind of candidates of the reference picture. In picture identifying information of each individual list, a type of a reference picture of a picture identified by that picture identifying information is set. Further, used_by_curr is set for each piece of the picture identifying information of the RefPicSetStCurrBefore list, the RefPicSetStCurrAfter list, and the RefPicSetLtCurr list.

(178) In Step S72, the list creation section **88** creates a current picture list in which the picture identifying information of the current picture for which the LTRP is set as a type of the reference picture is registered.

(179) In Step S73, the list creation section **88** creates two temporal lists, i.e., a RefpicListTemp0 list and a RefpicListTemp1 list on the basis of the picture identifying information of the individual list and the used_by_curr.

(180) In Step S74, the list creation section **88** rearranges the order of the pieces of picture identifying information in the temporal lists, on the basis of ref pic list modification that is disposed in the slice header and specifies the order of the pieces of picture identifying information in the

temporal lists.

(181) In Step S75, on the basis of each of the temporal lists, the list creation section **88** creates a reference picture list in which a predetermined number of pieces of picture identifying information from the head, which are registered in the temporal list, are registered in order from the head.

(182) In Step S76, the list creation section **88** registers the picture identifying information of the current picture, which is registered in the current picture list, in the head of the reference picture list. The processing then returns to Step S33 of FIG. 16 and proceeds to Step S34.

(183) FIG. 19 is a flowchart for describing details of the predicted vector list generation processing in Step S37 of FIG. 16. The predicted vector list generation processing is performed for each of the reference picture lists in which the reference picture is registered.

(184) In Step S91 of FIG. 19, the predicted vector generation section **90** determines a predetermined candidate block to be a candidate block to be processed. In the first processing of Step S91, a predetermined spatial-direction candidate block is determined to be a candidate block to be processed.

(185) In Step S92, on the basis of the index supplied from the motion prediction/compensation section **89**, the predicted vector generation section **90** reads a type of the reference picture of the current block corresponding to that index from the reference picture list and holds the type. The predicted vector generation section **90** then determines whether the read type of the reference picture of the current block is the STRP or not.

(186) In the case where it is determined in Step S92 that the type of the reference picture of the current block is the STRP, the processing proceeds to Step S93. In Step S93, the predicted vector generation section **90** determines whether the type of the reference picture of the candidate block to be processed, which is held therein, is the STRP or not.

(187) In the case where it is determined in Step S93 that the type of the reference picture of the candidate block to be processed is the STRP, that is, in the case where the type of the reference picture of the current block and the type of the reference picture of the candidate block are both the STRP, the processing proceeds to Step S94.

(188) In Step S94, the predicted vector generation section **90** sets the candidate block to be processed to available. In Step S95, on the basis of a difference in POC between the reference picture of the current block and the reference picture of the candidate block, the predicted vector generation section **90** performs scaling on the candidate of the reference motion vector, which is the motion vector of the candidate block to be processed, and generates a predicted vector. The predicted vector generation section **90** then registers that predicted vector in the predicted vector list and advances the processing to Step S100.

(189) Meanwhile, in the case where it is determined in Step S92 that the type of the reference picture of the current block is not the STRP, that is, in the case where the type of the reference picture of the current block is the LTRP, the processing proceeds to Step S96.

(190) In Step S96, similarly to the processing in Step S93, the predicted vector generation section **90** determines whether the type of the reference picture of the candidate block to be processed is the STRP or not. In the case where it is determined in Step S96 that the type of the reference picture of the candidate block to be processed is the STRP, that is, in the case where the type of the reference picture of the current block is the LTRP and the type of the reference picture of the candidate block is the STRP, the processing proceeds to Step S97.

(191) In Step S97, the predicted vector generation section **90** sets the candidate block to be processed to not available and advances the processing to Step S100.

(192) Meanwhile, in the case where it is determined in Step S93 that the type of the reference picture of the candidate block to be processed is not the STRP, that is, in the case where the type of the reference picture of the current block is the STRP and the type of the reference picture of the candidate block is the LTRP, the processing proceeds to Step S97. Thus, the candidate block to be processed is set to not available.

(193) Further, in the case where it is determined in Step S96 that the type of the reference picture of the candidate block to be processed is not the STRP, that is, in the case where the type of the reference picture of the current block and the type of the reference picture of the candidate block are both the LTRP, the processing proceeds to Step S98.

(194) In Step S98, the predicted vector generation section 90 sets the candidate block to be processed to available. In Step S99, the predicted vector generation section 90 generates a candidate of the reference motion vector, which is the motion vector of the candidate block to be processed, to be a predicted vector without change. The predicted vector generation section 90 then registers that predicted vector in the predicted vector list and advances the processing to Step S100.

(195) In Step S100, the predicted vector generation section 90 determines whether a predetermined number of predicted vectors are registered in the predicted vector list or not. In the case where it is determined in Step S100 that a predetermined number of predicted vectors are not yet registered, the processing returns to Step S91. A predetermined spatial-direction candidate block or temporal-direction candidate block, which is not yet determined to be a candidate block to be processed, is determined to be a candidate block to be processed, and the subsequent processing is performed.

(196) Meanwhile, in the case where it is determined in Step S100 that a predetermined number of predicted vectors are registered, the predicted vector generation section 90 supplies the predicted vector list to the difference vector generation section 91. The processing then returns to Step S37 of FIG. 16 and proceeds to Step S38.

(197) As described above, at the time of encoding of a current motion vector used in Intra BC, in the case where the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block are different from each other, the encoding apparatus 50 sets the candidate block to not available. Thus, the motion vector of the candidate block, the actual type of the reference picture of which is different from the actual type of the reference picture of the current block, is not used for generation of the predicted vector. Therefore, the difference vector becomes small, and coding efficiency is improved.

(198) Further, the encoding apparatus 50 sets the type of the reference picture of the current block, the actual type of which is "Intra BC", to the LTRP. Thus, in the case where the actual type of the reference picture of the candidate block is the STRP, the encoding apparatus 50 sets the candidate block to not available. Therefore, it is possible to improve coding efficiency in the case where Intra BC and inter coding are communalized, without largely changing HEVC in the case where Intra BC and inter coding are not communalized.

(199) Moreover, the encoding apparatus 50 registers the picture identifying information of the current picture in a predetermined position of the reference picture list of the current block. Therefore, a decoding apparatus that will be described later can know whether the reference picture is a current picture or not on the basis of an index corresponding to a position, in the reference picture list, of the picture identifying information of the reference picture. As a result, it is possible to prevent useless access to the DRAM from occurring in the case where the reference picture is the current picture.

(200) (Configuration Example of First Embodiment of Decoding Apparatus)

(201) FIG. 20 is a block diagram showing a configuration example of a first embodiment of a decoding apparatus as an image processing apparatus to which the present disclosure is applied, the decoding apparatus decoding the encoded stream transmitted from the encoding apparatus 50 of FIG. 13.

(202) The decoding apparatus 110 of FIG. 20 is constituted of a reception section 111, an extraction section 112, and a decoding section 113.

(203) The reception section 111 of the decoding apparatus 110 receives the encoded stream transmitted from the encoding apparatus 50 of FIG. 13 and supplies the encoded stream to the extraction section 112.

(204) The extraction section 112 extracts the parameter set and the encoded data from the encoded

stream supplied from the reception section **111** and supplies the parameter set and the encoded data to the decoding section **113**.

(205) The decoding section **113** decodes the encoded data, which is supplied from the extraction section **112**, by a system according to HEVC. At that time, the decoding section **113** also refers to the parameter set supplied from the extraction section **112**, as needed. The decoding section **113** outputs a picture obtained as a result of the decoding.

(206) (Configuration Example of Decoding Section)

(207) FIG. **21** is a block diagram showing a configuration example of the decoding section **113** of FIG. **20**.

(208) The decoding section **113** of FIG. **21** includes an accumulation buffer **131**, a lossless decoding section **132**, an inverse quantization section **133**, an inverse orthogonal transformation section **134**, an adder section **135**, a filter **136**, and a screen sorting buffer **139**. Further, the decoding section **113** includes a D/A conversion section **140**, a frame memory **141**, a switch **142**, an intra prediction section **143**, a predicted vector generation section **144**, a motion vector generation section **145**, a list creation section **146**, a motion compensation section **147**, and a switch **148**.

(209) The accumulation buffer **131** of the decoding section **113** receives the encoded data from the extraction section **112** of FIG. **20** and accumulates the encoded data. The accumulation buffer **131** supplies the accumulated encoded data, as the encoded data of the current picture, to the lossless decoding section **132**.

(210) The lossless decoding section **132** performs lossless decoding, such as variable-length decoding and arithmetic decoding, which corresponds to the lossless encoding of the lossless encoding section **76** of FIG. **14**, on the encoded data from the accumulation buffer **131**, to acquire the quantized orthogonal transformation coefficient and the encoding information. The lossless decoding section **132** supplies the quantized orthogonal transformation coefficient to the inverse quantization section **133**. Further, the lossless decoding section **132** supplies the intra prediction mode information and the like, as the encoding information, to the intra prediction section **143**. The lossless decoding section **132** supplies the index of the reference picture to the predicted vector generation section **144**, supplies the motion vector information to the motion vector generation section **145**, and supplies the inter prediction mode information and the index of the reference picture to the motion compensation section **147**.

(211) Moreover, the lossless decoding section **132** supplies the intra prediction mode information or inter prediction mode information, as the encoding information, to the switch **148**. The lossless decoding section **132** supplies the offset filter information, as the encoding information, to the filter **136**.

(212) The inverse quantization section **133**, the inverse orthogonal transformation section **134**, the adder section **135**, the filter **136**, the frame memory **141**, the switch **142**, the intra prediction section **143**, the predicted vector generation section **144**, the list creation section **146**, and the motion compensation section **147** respectively perform processing similar to the inverse quantization section **79**, the inverse orthogonal transformation section **80**, the adder section **81**, the filter **82**, the frame memory **85**, the switch **86**, the intra prediction section **87**, the predicted vector generation section **90**, the list creation section **88**, and the motion prediction/compensation section **89**. Thus, the picture is decoded.

(213) Specifically, the inverse quantization section **133** inversely quantizes the quantized orthogonal transformation coefficient supplied from the lossless decoding section **132** and supplies the resultant orthogonal transformation coefficient to the inverse orthogonal transformation section **134**.

(214) The inverse orthogonal transformation section **134** performs inverse orthogonal transformation on the orthogonal transformation coefficient supplied from the inverse quantization section **133** in units of TU. The inverse orthogonal transformation section **134** supplies residual

information, which is obtained as a result of the inverse orthogonal transformation, to the adder section **135**.

(215) The adder section **135** adds the residual information supplied from the inverse orthogonal transformation section **134** and a predicted image supplied from the switch **148**, to locally decode the current picture. It should be noted that in the case where the predicted image is not supplied from the switch **148**, the adder section **135** sets the residual information, which is supplied from the inverse orthogonal transformation section **134**, to a decoding result. The adder section **135** supplies the locally-decoded current picture, which is obtained as a result of the decoding, to the frame memory **141**. Further, the adder section **135** supplies the current picture, all regions of which are decoded, as a decoded picture, to the filter **136**.

(216) The filter **136** performs filter processing on the decoded picture supplied from the adder section **135**. Specifically, the filter **136** first performs deblocking filter processing on the decoded picture. Next, the filter **136** performs adaptive offset filter processing, a kind of which is indicated by the offset filter information, on the decoded picture after subjected to the deblocking filter processing for each of the LCUs by using an offset indicated by the offset filter information from the lossless decoding section **132**. The filter **136** supplies the decoded picture after subjected to the adaptive offset filter processing to the frame memory **141** and the screen sorting buffer **139**.

(217) The screen sorting buffer **139** stores, in units of frame, the decoded pictures supplied from the filter **136**. The screen sorting buffer **139** sorts the decoded pictures in units of frame, which are stored in encoding order, in the original display order, and supplies those pictures to the D/A conversion section **140**.

(218) The D/A conversion section **140** performs D/A conversion on the decoded pictures in units of frame, which are supplied from the screen sorting buffer **139**, and outputs those pictures.

(219) The frame memory **141** is constituted of, for example, a cache and a DRAM. The frame memory **141** stores the current picture, which is supplied from the adder section **135**, in the cache and stores the decoded picture, which is supplied from the filter **136**, in the DRAM. Pixels adjacent to the current block in the current picture stored in the cache are set as peripheral pixels and supplied to the intra prediction section **143** via the switch **142**.

(220) Further, the current picture stored in the cache or the decoded picture stored in the DRAM is set as a reference picture and output to the motion compensation section **147** via the switch **142**.

(221) The intra prediction section **143** performs the intra prediction processing of the optimal intra prediction mode, which is indicated by the intra prediction mode information supplied from the lossless decoding section **132**, on the current block by using the peripheral pixels read from the frame memory **141** via the switch **142**. The intra prediction section **143** supplies the resultant predicted image to the switch **148**.

(222) The predicted vector generation section **144** generates a predicted vector list of the current motion vector on the basis of the index supplied from the lossless decoding section **132** and the reference picture list held in the list creation section **146**, similarly to the predicted vector generation section **90** of FIG. **14**. The predicted vector generation section **144** supplies the predicted vector list to the motion vector generation section **145**.

(223) On the basis of the index in the motion vector information supplied from the lossless decoding section **132**, the motion vector generation section **145** reads a predicted vector corresponding to that index from the predicted vector list supplied from the predicted vector generation section **144**. The motion vector generation section **145** adds the predicted vector and a difference vector in the motion vector information, to perform prediction decoding on the current motion vector. The motion vector generation section **145** supplies the resultant current motion vector to the motion compensation section **147**.

(224) Similarly to the list creation section **88** of FIG. **14**, the list creation section **146** performs the reference picture list creation processing on the basis of the reference picture list information included in the SPS extracted by the extraction section **112** of FIG. **20**, the information disposed in

the slice header, and the like. The list creation section **146** holds and supplies the resultant reference picture list to the motion compensation section **147**.

(225) The motion compensation section **147** performs motion compensation processing on the current block on the basis of the inter prediction mode information from the lossless decoding section **132**, the index of the reference picture, and the current motion vector from the motion vector generation section **145**.

(226) Specifically, the motion compensation section **147** reads the picture identifying information of the index of the reference picture, in the pieces of the picture identifying information registered in the picture list held in the list creation section **146**. The motion compensation section **147** reads a reference picture identified by the read picture identifying information from the frame memory **141** via the switch **142**. The motion compensation section **147** performs motion compensation processing on the current block in the optimal inter prediction mode indicated by the inter prediction mode information, by using the reference picture and the current motion vector. The motion compensation section **147** supplies the resultant predicted image to the switch **148**.

(227) In the case where the intra prediction mode information is supplied from the lossless decoding section **132**, the switch **148** supplies the predicted image, which is supplied from the intra prediction section **143**, to the adder section **135**. Meanwhile, in the case where the inter prediction mode information is supplied from the lossless decoding section **132**, the switch **148** supplies the predicted image, which is supplied from the motion compensation section **147**, to the adder section **135**.

(228) (Description on Processing of Decoding Apparatus)

(229) FIG. **22** is a flowchart for describing image generation processing of the decoding apparatus **110** of FIG. **20**.

(230) In Step **S111** of FIG. **22**, the reception section **111** of the decoding apparatus **110** receives the encoded stream transmitted from the encoding apparatus **50** of FIG. **13** and supplies the encoded stream to the extraction section **112**.

(231) In Step **S112**, the extraction section **112** extracts the encoded data and the parameter set from the encoded stream supplied from the reception section **111** and supplies the encoded data and the parameter set to the decoding section **113**.

(232) In Step **S113**, the decoding section **113** performs decoding processing of decoding the encoded data, which is supplied from the extraction section **112**, by a system according to HEVC by using as needed the parameter set supplied from the extraction section **112**. Details of the decoding processing will be described with reference to FIG. **23** that will be described later. The processing is thus terminated.

(233) FIG. **23** is a flowchart for describing details of the decoding processing in Step **S113** of FIG. **22**.

(234) In Step **S131** of FIG. **23**, the accumulation buffer **131** (FIG. **21**) of the decoding section **113** receives the encoded data in units of frame from the extraction section **112** and accumulates the encoded data. The accumulation buffer **131** supplies the accumulated encoded data, as the encoded data of the current picture, to the lossless decoding section **132**.

(235) In Step **S132**, the lossless decoding section **132** performs lossless decoding on the encoded data from the accumulation buffer **131** and acquires the quantized orthogonal transformation coefficient and the encoding information. The lossless decoding section **132** supplies the quantized orthogonal transformation coefficient to the inverse quantization section **133**.

(236) Further, the lossless decoding section **132** supplies the intra prediction mode information and the like, as the encoding information, to the intra prediction section **143**. The lossless decoding section **132** supplies the index of the reference picture to the predicted vector generation section **144**, supplies the motion vector information to the motion vector generation section **145**, and supplies the inter prediction mode information and the index of the reference picture to the motion compensation section **147**.

(237) Moreover, the lossless decoding section **132** supplies the intra prediction mode information or inter prediction mode information, as the encoding information, to the switch **148**. The lossless decoding section **132** supplies the offset filter information, as the encoding information, to the filter **136**.

(238) In Step **S133**, the list creation section **146** performs the reference picture list creation processing, which is similar to the reference picture creation processing of FIG. **18**, on the basis of the reference picture list information included in the SPS extracted by the extraction section **112**, the information disposed in the slice header, and the like.

(239) In Step **S134**, the inverse quantization section **133** inversely quantizes the quantized orthogonal transformation coefficient supplied from the lossless decoding section **132** and supplies the resultant orthogonal transformation coefficient to the inverse orthogonal transformation section **134**.

(240) In Step **S135**, the inverse orthogonal transformation section **134** performs inverse orthogonal transformation on the orthogonal transformation coefficient supplied from the inverse quantization section **133** and supplies the resultant residual information to the adder section **135**.

(241) In Step **S136**, the motion compensation section **147** determines whether the inter prediction mode information is supplied from the lossless decoding section **132** or not. In the case where it is determined in Step **S136** that the inter prediction mode information is supplied, the processing proceeds to Step **S137**.

(242) In Step **S137**, the predicted vector generation section **144** performs the predicted vector list generation processing, which is similar to the predicted vector list generation processing of FIG. **19**, on the basis of the index supplied from the lossless decoding section **132** and the reference picture list held in the list creation section **146**.

(243) In Step **S138**, on the basis of the index in the motion vector information, the motion vector generation section **145** reads a predicted vector corresponding to that index from the predicted vector list. The motion vector generation section **145** adds the predicted vector and a difference vector in the motion vector information, to perform prediction decoding on the current motion vector. The motion vector generation section **145** supplies the resultant current motion vector to the motion compensation section **147**.

(244) In Step **S139**, the motion compensation section **147** performs the motion compensation processing on the current block on the basis of the inter prediction mode information and the index of the reference picture from the lossless decoding section **132**, and the current motion vector from the motion vector generation section **145**. The motion compensation section **147** supplies the resultant predicted image to the switch **148** and advances the processing to Step **S141**.

(245) Meanwhile, in the case where it is determined in Step **S136** that the inter prediction mode information is not supplied, that is, in the case where the intra prediction mode information is supplied to the intra prediction section **143**, the processing proceeds to Step **S140**.

(246) In Step **S140**, the intra prediction section **143** performs the intra prediction processing of the optimal intra prediction mode, which is indicated by the intra prediction mode information, on the current block by using the peripheral pixels read from the frame memory **141** via the switch **142**. The intra prediction section **143** supplies a predicted image generated as a result of the intra prediction processing to the adder section **135** via the switch **148** and advances the processing to Step **S141**.

(247) In Step **S141**, the adder section **135** adds the residual information supplied from the inverse orthogonal transformation section **134** and the predicted image supplied from the switch **148**, to locally decode the current picture. The adder section **135** supplies the locally-decoded current picture, which is obtained as a result of the decoding, to the frame memory **141**. Further, the adder section **135** supplies the current picture, all regions of which are decoded, as a decoded picture, to the filter **136**.

(248) In Step **S142**, the filter **136** performs deblocking filter processing on the decoded picture

supplied from the adder section **135**, to remove block distortion.

(249) In Step **S143**, the filter **136** performs adaptive offset filter processing on the decoded picture after subjected to the deblocking filter processing for each of the LCUs, on the basis of the offset filter information supplied from the lossless decoding section **132**. The filter **136** supplies the picture after subjected to the adaptive offset filter processing to the screen sorting buffer **139** and the frame memory **141**.

(250) In Step **S144**, the frame memory **141** stores the current picture, which is supplied from the adder section **81**, in the cache and stores the decoded picture, which is supplied from the filter **136**, in the DRAM. Pixels adjacent to the current block in the current picture stored in the cache are set as peripheral pixels and supplied to the intra prediction section **143** via the switch **142**. Further, the current picture stored in the cache or the decoded picture stored in the DRAM is set as a reference picture and output to the motion compensation section **147** via the switch **142**.

(251) In Step **S145**, the screen sorting buffer **139** stores, in units of frame, the decoded pictures supplied from the filter **136**, sorts the pictures in units of frame, which are stored in encoding order, in the original display order, and supplies those pictures to the D/A conversion section **140**.

(252) In Step **S146**, the D/A conversion section **140** performs D/A conversion on the pictures in units of frame, which are supplied from the screen sorting buffer **139**, and outputs the pictures. The processing then returns to Step **S113** of FIG. **22** and is terminated.

(253) As described above, at the time of decoding of a current motion vector used in Intra BC, in the case where the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block are different from each other, the decoding apparatus **110** sets the candidate block to not available. Thus, the motion vector of the candidate block, the actual type of the reference picture of which is different from the actual type of the reference picture of the current block, is not used for generation of the predicted vector.

(254) Therefore, the reference motion vector of the candidate block, the actual type of the reference picture of which is different from the actual type of the reference picture of the current block, is not used for generation of the predicted vector, and thus an encoded stream having improved coding efficiency can be decoded.

(255) Further, the decoding apparatus **110** sets the type of the reference picture of the current block, the actual type of which is “Intra BC”, to the LTRP. Thus, in the case where the actual type of the reference picture of the candidate block is the STRP, the decoding apparatus **110** sets the candidate block to not available. Therefore, it is possible to decode an encoded stream having improved coding efficiency in the case where Intra BC and inter coding are communalized, without largely changing HEVC in the case where Intra BC and inter coding are not communalized.

(256) Moreover, the decoding apparatus **110** registers the picture identifying information of the current picture in a predetermined position of the reference picture list of the current block.

Therefore, the decoding apparatus **110** can know whether the reference picture is a current picture or not on the basis of an index corresponding to a position, in the reference picture list, of the picture identifying information of the reference picture. As a result, it is possible to prevent useless access to the DRAM from occurring in the case where the reference picture is the current picture.

Second Embodiment

General Outline of Second Embodiment

(257) The general outline of a second embodiment will first be described with reference to FIGS. **24** to **27**.

(258) In the second embodiment, the coding efficiency in the case where Intra BC and inter coding are commonalized is improved not by setting the type of the current picture, which is the reference picture in Intra BC, to the LTRP but by changing a method of generating a predicted vector list in the case where the actual type of the reference picture is “Intra BC”.

(259) FIGS. **24** to **26** are diagrams for describing a method of generating a predicted vector list in the case where the actual type of the reference picture is “Intra BC” in the second embodiment.

(260) It should be noted that DiffPicOrderCnt (X,Y) represents a difference (X-Y) in POC between a picture X and a picture Y. Further, RefPicListX[refIdxLX] and LX[refIdxLX] (X=0 or X=1) each represent a reference picture identified by picture identifying information of an index refIdxLX of a reference picture of a current block, in picture identifying information registered in a reference picture list RefPicListX used for encoding of a slice including the current block.

(261) Further, CurrPic represents a current picture, and ColPic represents a picture of a temporal-direction candidate block. Moreover, xNbAK and yNbAK respectively represent a position in an x direction and a position in a y direction of a spatial-direction candidate block (hereinafter, called left candidate block) adjacent to the current block on the left, the upper left, or the lower left. xNbB.sub.K and yNbB.sub.K respectively represent a position in the x direction and a position in the y direction of a spatial-direction candidate block (hereinafter, upper candidate block) adjacent to the current block on the upper or the upper right.

(262) listCol[refIdxCol] represents a reference picture identified by picture identifying information of an index refIdxCol of a reference picture of a temporal-direction candidate block, in picture identifying information registered in a reference picture list RefPicListX used for encoding of a slice including the temporal-direction candidate block.

(263) As shown in FIG. 24, in the second embodiment, in the case where the candidate block to be processed is the left candidate block, when DiffPicOrderCnt (RefPicListX[refIdxLX], CurrPic)=0 and when DiffPicOrderCnt (RefPicListX[refIdxLX], CurrPic) and DiffPicOrderCnt (RefPicListX[RefIdxLX[xNbAx] [yNbAK]], CurrPic) are equal to each other, the candidate block to be processed is set to available.

(264) In other words, in the case where the reference picture of the current picture and the reference picture of the left candidate block are each the current picture, that is, in the case where the actual type of the reference picture of the current picture and the actual type of the reference picture of the left candidate block are “Intra BC”, the left candidate block is set to available. A candidate of a reference motion vector, which is a motion vector of the left candidate block, is then generated as a predicted vector without change.

(265) Similarly, in the second embodiment, as shown in FIG. 25, in the case where the candidate block to be processed is the upper candidate block, when DiffPicOrderCnt (RefPicListX[refIdxLX], CurrPic)=0 and when DiffPicOrderCnt (RefPicListX[refIdxLX], CurrPic) and DiffPicOrderCnt (RefPicListX[RefIdxLX[xNbB.sub.K] [yNbB.sub.K]], CurrPic) are equal to each other, the candidate block to be processed is set to available.

(266) In other words, in the case where the actual type of the reference picture of the current picture and the actual type of the reference picture of the upper candidate block are “Intra BC”, the upper candidate block is set to available. A candidate of a reference motion vector, which is a motion vector of the upper candidate block, is then generated as a predicted vector without change. The generation of the predicted vector is performed for each of the reference picture lists in which the reference picture is registered.

(267) Further, in the second embodiment, as shown in FIG. 26, in the case where the candidate block to be processed is the temporal-direction candidate block, when DiffPicOrderCnt (LX[refIdxLX], CurrPic)=0 and when DiffPicOrderCnt (LX[refIdxLX], CurrPic) and DiffPicOrderCnt (listCol[refIdxCol], ColPic) for colPb are equal to each other, the candidate block to be processed is set to available.

(268) In other words, in the case where the actual type of the reference picture of the current picture and the actual type of the reference picture of the temporal-direction candidate block are “Intra BC”, the temporal-direction candidate block is set to available. A candidate of a reference motion vector, which is a motion vector of the temporal-direction candidate block, is then generated as a predicted vector without change.

(269) As a result, as shown in FIG. 27, in the case where the actual type of the reference picture of the current picture is “Intra BC”, and only in the case where the actual type of the reference picture

of the candidate block is “Intra BC”, the motion vector of the candidate block is generated as a predicted vector without change. Meanwhile, in the cases other than above, the candidate block is set to not available. It should be noted that the type of the candidate of the reference picture, the actual type of which is “Intra BC”, is set to STRP.

(270) Configurations of an encoding apparatus and a decoding apparatus in the second embodiment are similar to those of the encoding apparatus **50** of FIG. **13** and the decoding apparatus **110** of FIG. **20**, except for the reference picture list creation processing performed in the list creation section **88** and the list creation section **146** and the predicted vector list generation processing performed in the predicted vector generation section **90** and the predicted vector generation section **144**. Therefore, hereinafter, the sections of the encoding apparatus **50** of FIG. **13** and the decoding apparatus **110** of FIG. **20** will be used as the sections of the encoding apparatus and the decoding apparatus in the second embodiment, and only the reference picture list creation processing and the predicted vector generation processing will be described.

(271) (Description on Processing of Encoding Apparatus)

(272) FIG. **28** is a flowchart for describing the reference picture list creation processing of the list creation section **88** of the encoding apparatus **50** in the second embodiment.

(273) In Step **S161** of FIG. **28**, the list creation section **88** creates four individual lists similarly to the processing of Step **S71** in FIG. **18**. STRP or LTRP is set for picture identifying information of each of the individual lists.

(274) In Step **S162**, the list creation section **88** creates a current picture list in which the picture identifying information of the current picture for which the STRP is set as a type of the reference picture is registered.

(275) Processing in Steps **S163** to **S166** are similar to the processing in Steps **S73** to **S76** of FIG. **18**, and description thereof will thus be omitted.

(276) It should be noted that, although not shown in the figures, the reference picture list creation processing of the list creation section **146** of the decoding apparatus **110** in the second embodiment is also performed similarly to the reference picture list creation processing of FIG. **28**.

(277) FIG. **29** is a flowchart for describing details of the predicted vector list generation processing of the predicted vector generation section **90** of the encoding apparatus **50** in the second embodiment. The predicted vector list generation processing is performed for each of the reference picture lists in which the reference picture is registered.

(278) Processing in Steps **S181** to **S183** of FIG. **29** are similar to the processing in Steps **S91** to **S93** of FIG. **19**, and description thereof will thus be omitted. In the case where it is determined in Step **S183** that the type of the reference picture of the candidate block to be processed is the STRP, the processing proceeds to Step **S184**. In Step **S184**, the predicted vector generation section **90** determines whether both the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block are the STRP or not. Specifically, for example, the predicted vector generation section **90** determines whether a difference in POC between the reference picture of the current block and the current picture and a difference in POC between the reference picture of the candidate block and the picture of the candidate block are both 1 or more.

(279) In the case where the predicted vector generation section **90** determines that the difference in POC between the reference picture of the current block and the current picture and the difference in POC between the reference picture of the candidate block and the picture of the candidate block are both 1 or more, the predicted vector generation section **90** determines that both the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block are the STRP, and advances the processing to Step **S185**.

(280) Processing in Steps **S185** and **S186** are similar to the processing in Steps **S94** and **S95** of FIG. **19**, and description thereof will thus be omitted.

(281) Meanwhile, in the case where at least one of the difference in POC between the reference picture of the current block and the current picture and the difference in POC between the reference

picture of the candidate block and the picture of the candidate block is 0, the predicted vector generation section **90** determines in Step **S184** that at least one of the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block is not the STRP.

(282) In Step **S187**, the predicted vector generation section **90** determines whether both the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block are “Intra BC” or not. Specifically, the predicted vector generation section **90** determines whether the difference in POC between the reference picture of the current block and the current picture and the POC between the reference picture of the candidate block and the picture of the candidate block are 0 or not.

(283) In the case where the predicted vector generation section **90** determines that the difference in POC between the reference picture of the current block and the current picture and or the POC between the reference picture of the candidate block and the picture of the candidate block is not 0, the predicted vector generation section **90** determines that any one of the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block is not “Intra BC”, and advances the processing to Step **S188**.

(284) In other words, in the case where the reference picture of the current block is the current picture but the picture of the candidate block and the reference picture of the candidate block are different from each other, and in the case where the picture of the candidate block and the reference picture of the candidate block are the same but the reference picture of the current block is not the current picture, the predicted vector generation section **90** determines that the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block are different from each other, and advances the processing to Step **S188**.

(285) In Step **S188**, the predicted vector generation section **90** sets the candidate block to be processed to not available.

(286) Further, in the case where it is determined in Step **S182** that the type of the reference picture of the current block is not the STRP, that is, in the case where the type of the reference picture of the current block is the LTRP, the processing proceeds to Step **S189**.

(287) In Step **S189**, similarly to the processing of Step **S183**, the predicted vector generation section **90** determines whether the type of the reference picture of the candidate block to be processed is the STRP or not. In the case where it is determined in Step **S189** that the type of the reference picture of the candidate block to be processed is the STRP, that is, in the case where the type of the reference picture of the current block is the LTRP and the type of the reference picture of the candidate block is the STRP, the processing proceeds to Step **S188**. Thus, the candidate block to be processed is set to not available.

(288) Meanwhile, in the case where it is determined in Step **S189** that the type of the reference picture of the candidate block to be processed is not the STRP, that is, in the case where both the type of the reference picture of the current block and the type of the reference picture of the candidate block are the LTRP, the processing proceeds to Step **S190**. Processing in Steps **S190** and **S191** are similar to the processing in Steps **S98** and **S99** of FIG. **19**, and description thereof will thus be omitted.

(289) Meanwhile, in the case where the predicted vector generation section **90** determines that the difference in POC between the reference picture of the current block and the current picture and the difference in POC between the reference picture of the candidate block and the picture of the candidate block are both 0, the predicted vector generation section **90** determines in Step **S187** that both the actual type of the reference picture of the current block and the actual type of the reference picture of the candidate block are “Intra BC”.

(290) The predicted vector generation section **90** then advances the processing to Step **S190**. Thus, the candidate block to be processed is set to available, and the candidate of the reference motion vector of the candidate block to be processed is generated as a predicted vector. After the

processing of Step **S186**, **S188**, or

(291) **S191**, the processing proceeds to Step **S192**. The processing of Step **S192** is similar to the processing of Step **S100** in FIG. **19**, and description thereof will thus be omitted.

(292) It should be noted that, although not shown in the figures, the predicted vector list generation processing of the predicted vector generation section **144** of the decoding apparatus **110** in the second embodiment is also performed similarly to the predicted vector list generation processing of FIG. **29**.

Third Embodiment

General Outline of Third Embodiment

(293) In a third embodiment, the picture identifying information of the current picture is registered in the head of the temporal list, not in the reference picture list.

(294) In other words, in the third embodiment, as shown in FIG. **30**, after the two temporal lists, i.e., the RefpicListTemp0 list and the RefpicListTemp1 list, are created on the basis of the individual lists of FIG. **10**, the picture identifying information of the current picture is registered in the head of each of the temporal lists.

(295) Next, the order of the pieces of the picture identifying information registered in the RefpicListTemp0 list and the RefpicListTemp1 list is changed as needed (Reference reordering). However, the order of the picture identifying information of the current picture is prohibited from being changed.

(296) Lastly, a reference picture list RefpicList0 is created, in which the picture identifying information in the head, and the pieces of the picture identifying information in number (five in the example of FIG. **30**) obtained by adding 1 to num_ref_idx_10_active_minus1, which are registered in the RefpicListTemp0 list, are registered in order from the head.

(297) Further, a reference picture list RefpicList1 is created, in which the picture identifying information in the head, and the pieces of the picture identifying information in number (four in the example of FIG. **30**) obtained by adding 1 to num_ref_idx_11_active_minus1, which are registered in the RefpicListTemp1 list, are registered in order from the head.

(298) Configurations of an encoding apparatus and a decoding apparatus in the third embodiment are similar to those of the encoding apparatus **50** of FIG. **13** and the decoding apparatus **110** of FIG. **20**, except for the reference picture list creation processing performed in the list creation section **88** and the list creation section **146**. Therefore, hereinafter, the sections of the encoding apparatus **50** of FIG. **13** and the decoding apparatus **110** of FIG. **20** will be used as the sections of the encoding apparatus and the decoding apparatus in the third embodiment, and only the reference picture list creation processing will be described.

(299) (Description on Processing of Encoding Apparatus)

(300) FIG. **31** is a flowchart for describing the reference picture list creation processing of the list creation section **88** of the encoding apparatus **50** in the third embodiment.

(301) Processing in Steps **S201** to **S203** of FIG. **31** are similar to the processing in Steps **S71** to **S73** of FIG. **18**, and description thereof will thus be omitted.

(302) In Step **S204**, the list creation section **88** registers the picture identifying information of the current picture, which is registered in the current picture list, in the head of the temporal list.

(303) In Step **S205**, the list creation section **88** rearranges the order of the pieces of picture identifying information in the temporal lists, other than the picture identifying information of the current picture, on the basis of ref_pic_list_modification that is disposed in the slice header.

(304) In Step **S206**, the list creation section **88** creates a reference picture list on the basis of each of the temporal lists. In other words, the list creation section **88** creates a reference picture list in which the picture identifying information in the head, and the pieces of the picture identifying information in number obtained by adding 1 to num_ref_idx_10_active_minus1

(num_ref_idx_11_active_minus1) disposed in the slice header, which are registered in the temporal list, are registered in order from the head.

(305) It should be noted that, although not shown in the figures, the reference picture list creation processing of the list creation section **146** of the decoding apparatus **110** in the third embodiment is also performed similarly to the reference picture list creation processing of FIG. **31**.

Fourth Embodiment

(306) (Description on Computer to which Present Disclosure is Applied)

(307) The series of processing described above can be executed by hardware or software. In the case where the series of processing is executed by software, a program configuring the software is installed in a computer. Here, the computer includes a computer incorporated in dedicated hardware and, for example, a general-purpose personal computer that can execute various functions by installing various programs therein.

(308) FIG. **32** is a block diagram showing a hardware configuration example of a computer that executes the series of processing described above by a program.

(309) In the computer, a CPU (Central Processing Unit) **201**, a ROM (Read Only Memory) **202**, and a RAM (Random Access Memory) **203** are connected to one another via a bus **204**.

(310) Moreover, an input/output interface **205** is connected to the bus **204**. An input section **206**, an output section **207**, a storage section **208**, a communication section **209**, and a drive **210** are connected to the input/output interface **205**.

(311) The input section **206** is constituted of a keyboard, a mouse, a microphone, and the like. The output section **207** is constituted of a display, a speaker, and the like. The storage section **208** is constituted of a hard disk, a nonvolatile memory, and the like. The communication section **209** is constituted of a network interface and the like. The drive **210** drives a removable medium **211** such as a magnetic disk, an optical disc, a magneto-optical disk, or a semiconductor memory.

(312) In the computer configured as described above, the CPU **201** loads the program stored in, for example, the storage section **208** to the RAM **203** via the input/output interface **205** and the bus **204** and executes the program, to perform the series of processing described above.

(313) The program executed by the computer (CPU **201**) can be provided by, for example, being recorded on the removable medium **211** as a package medium or the like. Further, the program can be provided via a wired or wireless transmission medium such as a local area network, the Internet, and digital satellite broadcasting.

(314) In the computer, when the removable medium **211** is mounted to the drive **210**, the program can be installed in the storage section **208** via the input/output interface **205**. Further, the program can be received by the communication section **209** via the wired or wireless transmission medium and installed in the storage section **208**. In addition, the program can be previously installed in the ROM **202** or the storage section **208**.

(315) It should be noted that the program executed by the computer may be a program that is processed chronologically along the described order in this specification or may be a program that is processed in parallel or at a necessary timing such as when an invocation is performed.

Fifth Embodiment

(316) (Configuration Example of Television Apparatus)

(317) FIG. **33** exemplifies a schematic configuration of a television apparatus to which the present disclosure is applied. A television apparatus **900** includes an antenna **901**, a tuner **902**, a demultiplexer **903**, a decoder **904**, a video signal processing section **905**, a display section **906**, an audio signal processing section **907**, a speaker **908**, and an external interface section **909**.

Moreover, the television apparatus **900** includes a control section **910**, a user interface section **911**, and the like.

(318) The tuner **902** selects a desired channel from broadcast wave signals received by the antenna **901** for demodulation, and outputs an encoded bit stream obtained to the demultiplexer **903**.

(319) The demultiplexer **903** extracts a video or audio packet of a program to be viewed from the encoded bit stream and outputs data of the extracted packet to the decoder **904**. Further, the demultiplexer **903** supplies a packet of data such as EPG (Electronic Program Guide) to the control

section **910**. It should be noted that in the case where scramble is performed, the scramble is released by the demultiplexer or the like.

(320) The decoder **904** performs packet decoding processing and outputs video data generated by the decoding processing to the video signal processing section **905** and audio data generated by the decoding processing to the audio signal processing section **907**.

(321) The video signal processing section **905** performs noise removal, video processing corresponding to user setting, or the like on the video data. The video signal processing section **905** generates video data of a program to be displayed on the display section **906**, image data processed on the basis of an application supplied via a network, and the like. Further, the video signal processing section **905** generates video data for displaying a menu screen to select items etc. and superimposes such video data on the video data of the program. The video signal processing section **905** generates a drive signal on the basis of the video data thus generated and drives the display section **906**.

(322) The display section **906** drives a display device (for example, liquid crystal display element) on the basis of the drive signal from the video signal processing section **905**, to display a video of the program or the like.

(323) The audio signal processing section **907** performs predetermined processing such as noise removal on the audio data, performs D/A conversion processing or amplification processing on the processed audio data, and supplies such audio data to the speaker **908**, to perform audio output.

(324) The external interface section **909** is an interface for connection with an external device or a network, and sends and receives data such as the video data and the audio data.

(325) The user interface section **911** is connected to the control section **910**. The user interface section **911** is constituted of an operation switch, a remote control signal reception section, and the like and supplies an operation signal corresponding to a user operation to the control section **910**.

(326) The control section **910** is constituted of a CPU (Central Processing Unit), a memory, and the like. The memory stores programs executed by the CPU, various types of data necessary for the CPU to perform processing, EPG data, data acquired via the network, and the like. The programs stored in the memory are read and executed by the CPU at a predetermined timing such as activation of the television apparatus **900**. The CPU executes the programs to control the sections such that the television apparatus **900** operates according to the user operation.

(327) It should be noted that in the television apparatus **900**, a bus **912** is provided to connect the tuner **902**, the demultiplexer **903**, the video signal processing section **905**, the audio signal processing section **907**, the external interface section **909**, and the like to the control section **910**.

(328) In the television apparatus thus configured, the decoder **904** is provided with a function of the decoding apparatus (decoding method) of the subject application. This makes it possible to decode an encoded stream having improved coding efficiency at the time when a prediction using a correlation within a screen is performed.

Sixth Embodiment

(329) (Configuration Example of Mobile Phone)

(330) FIG. **34** exemplifies a schematic configuration of a mobile phone to which the present disclosure is applied. A mobile phone **920** includes a communication section **922**, an audio codec **923**, a camera section **926**, an image processing section **927**, a multiplex/demultiplex section **928**, a recording/reproduction section **929**, a display section **930**, and a control section **931**. Those components are connected to one another via a bus **933**.

(331) Further, an antenna **921** is connected to the communication section **922**, and a speaker **924** and a microphone **925** are connected to the audio codec **923**. Furthermore, an operation section **932** is connected to the control section **931**.

(332) The mobile phone **920** performs various operations such as sending and reception of audio signals, sending and reception of e-mail or image data, imaging, and data recording, in various modes such as a voice communication mode and a data communication mode.

(333) In the voice communication mode, an audio signal generated in the microphone **925** is subjected to conversion into audio data or data compression in the audio codec **923** and then supplied to the communication section **922**. The communication section **922** performs modulation processing, frequency conversion processing, or the like on the audio data to generate a sending signal. Further, the communication section **922** supplies the sending signal to the antenna **921** to be sent to a base station not shown in the figure. Further, the communication section **922** performs amplification, frequency conversion processing, demodulation processing, or the like on a received signal received in the antenna **921** and supplies obtained audio data to the audio codec **923**. The audio codec **923** performs data decompression of the audio data or conversion into an analog audio signal, and outputs the resultant signal to the speaker **924**.

(334) Further, in the case where mail is sent in the data communication mode, the control section **931** receives character data input by an operation of the operation section **932** and displays the input characters on the display section **930**. Further, the control section **931** generates mail data on the basis of a user instruction or the like in the operation section **932** and supplies the mail data to the communication section **922**. The communication section **922** performs modulation processing, frequency conversion processing, or the like on the mail data and sends an obtained sending signal from the antenna **921**. Further, the communication section **922** performs amplification, frequency conversion processing, demodulation processing, or the like on a received signal received in the antenna **921**, to restore mail data. The mail data is supplied to the display section **930** to display mail content.

(335) It should be noted that the mobile phone **920** also enables the recording/reproduction section **929** to store the received mail data in a storage medium. The storage medium is an arbitrary rewritable storage medium. For example, the storage medium is a semiconductor memory such as a RAM and a built-in flash memory, a hard disk, or a removable medium such as a magnetic disk, a magneto-optical disk, an optical disc, a USB (Universal Serial Bus) memory, and a memory card.

(336) In the case where image data is sent in the data communication mode, image data generated in the camera section **926** is supplied to the image processing section **927**. The image processing section **927** performs encoding processing on the image data and generates encoded data.

(337) The multiplex/demultiplex section **928** multiplexes the encoded data generated in the image processing section **927** and audio data supplied from the audio codec **923** by a predetermined system and supplies the resultant data to the communication section **922**. The communication section **922** performs modulation processing, frequency conversion processing, or the like on the multiplexed data and sends an obtained sending signal from the antenna **921**. Further, the communication section **922** performs amplification, frequency conversion processing, demodulation processing, or the like on a received signal received in the antenna **921**, to restore the multiplexed data. The multiplexed data is supplied to the multiplex/demultiplex section **928**. The multiplex/demultiplex section **928** demultiplexes the multiplexed data and supplies encoded data to the image processing section **927** and audio data to the audio codec **923**. The image processing section **927** performs decoding processing on the encoded data to generate image data. The image data is supplied to the display section **930** to display a received image. The audio codec **923** converts the audio data into an analog audio signal and supplies the analog audio signal to the speaker **924**, to output a received audio.

(338) In the mobile phone device thus configured, the image processing section **927** is provided with a function of the encoding apparatus and the decoding apparatus (encoding method and decoding method) of the subject application. This makes it possible to improve coding efficiency at the time when a prediction using a correlation within a screen is performed. Further, it is possible to decode an encoded stream having improved coding efficiency at the time when a prediction using a correlation within a screen is performed.

Seventh Embodiment

(339) (Configuration Example of Recording/Reproduction Apparatus)

(340) FIG. 35 exemplifies a schematic configuration of a recording/reproduction apparatus to which the present disclosure is applied. A recording/reproduction apparatus **940** records, for example, audio data and video data of a received broadcast program on a recording medium and provides the recorded data to a user at a timing corresponding to a user instruction. Further, the recording/reproduction apparatus **940** can also acquire, for example, audio data and video data from another apparatus and record those pieces of data on a recording medium. Moreover, the recording/reproduction apparatus **940** decodes and outputs the audio data and video data recorded on the recording medium, to enable a monitoring apparatus or the like to perform image display or audio output.

(341) The recording/reproduction apparatus **940** includes a tuner **941**, an external interface section **942**, an encoder **943**, an HDD (Hard Disk Drive) section **944**, a disk drive **945**, a selector **946**, a decoder **947**, an OSD (On-Screen Display) section **948**, a control section **949**, and a user interface section **950**.

(342) The tuner **941** selects a desired channel from broadcast wave signals received by an antenna not shown in the figure. The tuner **941** outputs an encoded bit stream, which is obtained by demodulation of a received signal of the desired channel, to the selector **946**.

(343) The external interface section **942** is constituted of at least any one of an IEEE1394 interface, a network interface section, a USB interface, a flash memory interface, and the like. The external interface section **942** is an interface for connection with an external device, a network, a memory card, and the like and receives data such as video data or audio data to be recorded.

(344) When the video data or audio data supplied from the external interface section **942** is not encoded, the encoder **943** performs encoding by a predetermined system and outputs an encoded bit stream to the selector **946**.

(345) The HDD section **944** records content data such as video and audio, various programs, and other data in a built-in hard disk and reads those pieces of data from the hard disk at the time of reproduction or the like.

(346) The disk drive **945** records and reproduces a signal with respect to a mounted optical disc. The optical disc is, for example, a DVD disc (DVD-Video, DVD-RAM, DVD-R, DVD-RW, DVD+R, DVD+RW, etc.) or a Blu-ray (registered trademark) disc.

(347) At the time of recording of video or audio, the selector **946** selects any encoded bit stream from the tuner **941** or the encoder **943** and supplies the encoded bit stream to any of the HDD section **944** and the disk drive **945**. Further, at the time of reproduction of video or audio, the selector **946** supplies an encoded bit stream output from the HDD section **944** or the disk drive **945** to the decoder **947**.

(348) The decoder **947** performs decoding processing on the encoded bit stream. The decoder **947** supplies video data generated by the decoding processing to the OSD section **948**. Further, the decoder **947** outputs audio data generated by the decoding processing.

(349) The OSD section **948** generates video data for displaying a menu screen to select items etc. and superimposes such video data on the video data output from the decoder **947** to be output.

(350) The user interface section **950** is connected to the control section **949**. The user interface section **950** is constituted of an operation switch, a remote control signal reception section, and the like and supplies an operation signal corresponding to a user operation to the control section **949**.

(351) The control section **949** is constituted of a CPU, a memory, and the like. The memory stores programs executed by the CPU and various types of data necessary for the CPU to perform processing. The programs stored in the memory are read and executed by the CPU at a predetermined timing such as activation of the recording/reproduction apparatus **940**. The CPU executes the programs to control the sections such that the recording/reproduction apparatus **940** operates according to the user operation.

(352) In the recording/reproduction apparatus thus configured, the encoder **943** is provided with a function of the encoding apparatus (encoding method) of the subject application, and the decoder

947 is provided with a function of the decoding apparatus (decoding method) of the subject application. This makes it possible to improve coding efficiency at the time when a prediction using a correlation within a screen is performed. Further, it is possible to decode an encoded stream having improved coding efficiency at the time when a prediction using a correlation within a screen is performed.

Eighth Embodiment

(353) (Configuration Example of Imaging Apparatus)

(354) FIG. **36** exemplifies a schematic configuration of an imaging apparatus to which the present disclosure is applied. An imaging apparatus **960** images a subject and then displays an image of the subject on a display section or records such image data on a recording medium.

(355) The imaging apparatus **960** includes an optical block **961**, an imaging section **962**, a camera signal processing section **963**, an image data processing section **964**, a display section **965**, an external interface section **966**, a memory section **967**, a media drive **968**, an OSD section **969**, and a control section **970**. Further, a user interface section **971** is connected to the control section **970**. Moreover, the image data processing section **964**, the external interface section **966**, the memory section **967**, the media drive **968**, the OSD section **969**, the control section **970**, and the like are connected to one another via a bus **972**.

(356) The optical block **961** is constituted of a focus lens, a diaphragm mechanism, and the like. The optical block **961** forms an optical image of the subject on an imaging surface of the imaging section **962**. The imaging section **962** is constituted of a CCD or CMOS image sensor, and generates an electrical signal corresponding to the optical image by opto-electronic conversion and supplies the electrical signal to the camera signal processing section **963**.

(357) The camera signal processing section **963** performs various types of camera signal processing, such as knee correction, gamma correction, and color correction, on the electrical signal supplied from the imaging section **962**. The camera signal processing section **963** supplies image data after subjected to the camera signal processing to the image data processing section **964**.

(358) The image data processing section **964** performs encoding processing on the image data supplied from the camera signal processing section **963**. The image data processing section **964** supplies encoded data generated by the encoding processing to the external interface section **966** or the media drive **968**. Further, the image data processing section **964** performs decoding processing on encoded data supplied from the external interface section **966** or the media drive **968**. The image data processing section **964** supplies image data generated by the decoding processing to the display section **965**. Further, the image data processing section **964** performs processing of supplying the image data, which is supplied from the camera signal processing section **963**, to the display section **965**, or superimposes display data acquired from the OSD section **969** on the image data and supplies such display data to the display section **965**.

(359) The OSD section **969** generates display data of a menu screen constituted of symbols, characters, and graphics, icons, and the like and outputs the display data to the image data processing section **964**.

(360) The external interface section **966** is constituted of, for example, a USB input and output terminal and is connected to a printer in the case where an image is printed. Further, a drive is connected to the external interface section **966** as needed, and a removable medium such as a magnetic disk and an optical disc is appropriately mounted so as to install a computer program read therefrom as needed. Moreover, the external interface section **966** includes a network interface connected to a predetermined network such as a LAN (local area network) and the Internet. The control section **970** can read the encoded data from the media drive **968**, for example, according to an instruction from the user interface section **971**, and can supply the encoded data from the external interface section **966** to another apparatus connected via a network. Further, the control section **970** can acquire, via the external interface section **966**, encoded data or image data supplied

from another apparatus via a network, and can supply the data to the image data processing section **964**.

(361) As a recording medium driven by the media drive **968**, for example, an arbitrary readable and writable removable medium such as a magnetic disk, a magneto-optical disk, an optical disc, and a semiconductor memory is used. Further, the recording medium may be of an arbitrary type as a removable medium, and may be a tape device, a disc, or a memory card. As a matter of course, the recording medium may be a non-contact IC (Integrated Circuit) card or the like.

(362) Further, the media drive **968** and the recording medium may be integrated and constituted of a non-transportable storage medium such as a built-in hard disk drive and an SSD (Solid State Drive).

(363) The control section **970** is constituted of a CPU. The memory section **967** stores programs executed by the control section **970**, various types of data necessary for the control section **970** to perform processing, and the like. The programs stored in the memory section **967** are read and executed by the control section **970** at a predetermined timing such as activation of the imaging apparatus **960**. The control section **970** executes the programs to control the sections such that the imaging apparatus **960** operates according to a user operation.

(364) In the imaging apparatus thus configured, the image data processing section **964** is provided with a function of the encoding apparatus and the decoding apparatus (encoding method and decoding method) of the subject application. This makes it possible to improve coding efficiency at the time when a prediction using a correlation within a screen is performed. Further, it is possible to decode an encoded stream having improved coding efficiency at the time when a prediction using a correlation within a screen is performed.

Ninth Embodiment

Other Examples of Embodiment

(365) The examples of the apparatuses, the systems, and the like to which the present disclosure is applied have been described hereinabove, but the present disclosure is not limited thereto and can be carried out as any other configurations to be mounted to an apparatus constituting the apparatuses or systems as described above, for example, a processor as a system LSI (Large Scale Integration) or the like, a module using a plurality of processors or the like, a unit using a plurality of modules or the like, a set obtained by further adding another function to the unit, or the like (i.e., configuration of a part of an apparatus).

(366) (Configuration Example of Video Set)

(367) An example of the case where the present disclosure is carried out as a set will be described with reference to FIG. **37**. FIG. **37** shows an example of a schematic configuration of a video set to which the present disclosure is applied.

(368) Multi-functionality of electronic apparatuses has been advanced in recent years, and in the case where a part of a configuration thereof is carried out by selling, provision, or the like in the development and manufacturing thereof, there have been frequently found not only a case where the part is carried out as a configuration having one function but also a case where the part is carried out as one set having a plurality of functions by combining a plurality of configurations having associated functions.

(369) A video set **1300** shown in FIG. **37** has such a multi-functionalized configuration and is a combination of a device having a function on image encoding and decoding (any one or both of them) with a device having another function associated with that function.

(370) As shown in FIG. **37**, the video set **1300** includes a module group including a video module **1311**, an external memory **1312**, a power management module **1313**, a front end module **1314**, and the like, and devices having associated functions, such as a connectivity **1321**, a camera **1322**, and a sensor **1323**.

(371) The module is assumed as a component in which several component-wise functions associated with one another are collected to have a cohesive function. A specific physical

configuration is arbitrary, but the configuration is assumed to be, for example, one obtained by disposing a plurality of processors having respective functions, electronic circuit elements such as a resistor and a capacitor, other devices, and the like on a wiring substrate or the like for integration. Further, it is also conceived that the module may be combined with another module, a processor, and the like, to be a new module.

(372) In the case of the example of FIG. 37, the video module **1311** is a combination of configurations having functions on image processing and includes an application processor, a video processor, a broadband modem **1333**, and an RF (Radio Frequency) module **1334**.

(373) The processor is one in which a configuration having a predetermined function is integrated in a semiconductor chip by an SoC (System On a Chip) and may be called, for example, a system LSI (Large Scale Integration). Such a configuration having a predetermined function may be a logic circuit (hardware configuration), a CPU, a ROM, a RAM, or the like, and a program (software configuration) executed by using those above, or may be a combination of both the configurations. For example, the processor may include a logic circuit, a CPU, a ROM, a RAM, and the like and may achieve a part of the functions by the logic circuit (hardware configuration) and the other functions by the program (software configuration) executed by the CPU.

(374) The application processor **1331** of FIG. 37 is a processor that executes an application on image processing. The application executed by the application processor **1331** can not only perform computing processing but also control configurations inside and outside the video module **1311**, e.g., the video processor **1332**, as needed in order to achieve a predetermined function.

(375) The video processor **1332** is a processor having a function on image encoding/decoding (one or both of them).

(376) The broadband modem **1333** is a processor (or module) that performs processing on wired or wireless (or both of them) broadband communication performed via broadband connections such as the Internet and a public telephone network. For example, the broadband modem **1333** performs, for example, digital modulation on data to be sent (digital signal) for conversion into an analog signal or demodulates a received analog signal for conversion into data (digital signal). For example, the broadband modem **1333** can perform digital modulation/demodulation on arbitrary information such as image data processed by the video processor **1332**, a stream in which image data is encoded, an application program, and setting data.

(377) The RF module **1334** is a module that performs frequency conversion, modulation and demodulation, amplification, filter processing, or the like on an RF (Radio Frequency) signal that is sent and received via an antenna. For example, the RF module **1334** performs frequency conversion or the like on a baseband signal generated by the broadband modem **1333** and generates an RF signal. Further, for example, the RF module **1334** performs frequency conversion or the like on an RF signal received via the front end module **1314** and generates a baseband signal.

(378) It should be noted that, as indicated by a dotted line **1341** in FIG. 37, the application processor **1331** and the video processor **1332** may be integrated and configured as one processor.

(379) The external memory **1312** is a module that is provided outside the video module **1311** and includes a storage device used by the video module **1311**. The storage device of the external memory **1312** may be achieved by any physical configuration. In general, since the storage device is frequently used for storing a large amount of data such as image data in units of frame, the storage device may be desirably achieved by, for example, a semiconductor memory that is relatively inexpensiveness and has a large capacity, such as a DRAM (Dynamic Random Access Memory). The power management module **1313** manages and

(380) controls power supply to the video module **1311** (configurations within the video module **1311**).

(381) The front end module **1314** is a module that provides a front end function (a circuit at a sending and reception end on the antenna side) to the RF module **1334**. As shown in FIG. 37, the front end module **1314** includes, for example, an antenna section **1351**, a filter **1352**, and an

amplification section **1353**.

(382) The antenna section **1351** includes an antenna that sends and receives a radio signal, and peripheral configurations thereof. The antenna section **1351** sends a signal supplied from the amplification section **1353** as a radio signal, and supplies a received radio signal to the filter **1352** as an electrical signal (RF signal). The filter **1352** performs filter processing or the like on the RF signal received via the antenna section **1351** and supplies the processed RF signal to the RF module **1334**. The amplification section **1353** amplifies the RF signal supplied from the RF module **1334** and supplies the resultant signal to the antenna section **1351**.

(383) The connectivity **1321** is a module having a function on external connection. A physical configuration of the connectivity **1321** is arbitrary. For example, the connectivity **1321** includes a configuration having a communication function other than communication standards corresponding to the broadband modem **1333**, an external input/output terminal, and the like.

(384) For example, the connectivity **1321** may include a module having a communication function according to wireless communication standards such as Bluetooth (registered trademark), IEEE 802.11 (for example, Wi-Fi (Wireless Fidelity, registered trademark)), NFC (Near Field Communication), and IrDA (InfraRed Data Association), an antenna that sends and receives a signal according to such standards, and the like. Further, for example, the connectivity **1321** may include a module having a communication function according to wired communication standards such as USB (Universal Serial Bus) and HDMI (registered trademark) (High-Definition Multimedia Interface), and a terminal according to such standards. Moreover, for example, the connectivity **1321** may have another data (signal) transmission function or the like, such as an analog input/output terminal.

(385) It should be noted that the connectivity **1321** may include a device as a data (signal) transmission destination. For example, the connectivity **1321** may include a drive (including not only a drive of a removable medium but also a hard disk, an SSD (Solid State Drive), a NAS (Network Attached Storage), and the like) that reads and writes data on a recording medium such as a magnetic disk, an optical disc, a magneto-optical disk, and a semiconductor memory. Further, the connectivity **1321** may include an image or audio output device (monitor, speaker, or the like).

(386) The camera **1322** is a module having a function of imaging a subject and acquiring image data of the subject. The image data acquired by imaging of the camera **1322** is, for example, supplied to the video processor **1332** and encoded.

(387) The sensor **1323** is a module having an arbitrary sensor function, e.g., an audio sensor, an ultrasonic sensor, an optical sensor, an illuminance sensor, an infrared sensor, an image sensor, a rotation sensor, an angular sensor, an angular velocity sensor, a velocity sensor, an acceleration sensor, an inclination sensor, a magnetic identification sensor, a shock sensor, and a temperature sensor. Data detected by the sensor **1323** is, for example, supplied to the application processor **1331** and used by an application or the like.

(388) The configuration described above as a module may be achieved as a processor and, conversely, the configuration described as a processor may be achieved as a module.

(389) In the video set **1300** configured as described above, the present disclosure can be applied to the video processor **1332** as will be described later. Therefore, the video set **1300** can be carried out as a set to which the present disclosure is applied.

(390) (Configuration Example of Video Processor)

(391) FIG. **38** shows an example of a schematic configuration of the video processor **1332** (FIG. **37**) to which the present disclosure is applied.

(392) In the case of the example of FIG. **38**, the video processor **1332** has a function of receiving input of a video signal and an audio signal and encoding those signals by a predetermined system and a function of decoding the encoded video data and audio data and reproducing and outputting the video signal and the audio signal.

(393) As shown in FIG. **38**, the video processor **1332** includes a video input processing section

1401, a first image zooming section **1402**, a second image zooming section **1403**, a video output processing section **1404**, a frame memory **1405**, and a memory control section **1406**. Further, the video processor **1332** includes an encoding/decoding engine **1407**, video ES (Elementary Stream) buffers **1408A** and **1408B**, and audio ES buffers **1409A** and **1409B**. Moreover, the video processor **1332** includes an audio encoder **1410**, an audio decoder **1411**, a multiplexing section (MUX (Multiplexer)) **1412**, a demultiplexing section (DMUX (Demultiplexer)) **1413**, and a stream buffer **1414**.

(394) The video input processing section **1401** acquires a video signal input from, e.g., the connectivity **1321** (FIG. 37), and converts the video signal into digital image data. The first image zooming section **1402** performs format conversion, image zooming processing, or the like on the image data. The second image zooming section **1403** performs image zooming processing on the image data in accordance with a format of a destination for output via the video output processing section **1404** or perform format conversion, image zooming processing, or the like similar to the first image zooming section **1402** on the image data. The video output processing section **1404** performs format conversion, conversion into an analog signal, or the like on the image data and outputs the data, as a reproduced video signal, to the connectivity **1321** (FIG. 37), for example.

(395) The frame memory **1405** is a memory for image data shared with the video input processing section **1401**, the first image zooming section **1402**, the second image zooming section **1403**, the video output processing section **1404**, and the encoding/decoding engine **1407**. The frame memory **1405** is achieved as a semiconductor memory, e.g., a DRAM.

(396) In response to a synchronization signal from the encoding/decoding engine **1407**, the memory control section **1406** controls access for write and read with respect to the frame memory **1405** according to a schedule of access to the frame memory **1405**, which is written in an access management table **1406A**. The access management table **1406A** is updated by the memory control section **1406** in accordance with processing executed in the encoding/decoding engine **1407**, the first image zooming section **1402**, the second image zooming section **1403**, and the like.

(397) The encoding/decoding engine **1407** performs encoding processing on image data and decoding processing on a video stream in which the image data is encoded data. For example, the encoding/decoding engine **1407** encodes image data read from the frame memory **1405** and sequentially writes the image data, as a video stream, in the video ES buffer **1408A**. Further, for example, the encoding/decoding engine **1407** sequentially reads the video stream from the video ES buffer **1408B** for decoding and sequentially writes the decoded video stream, as image data, in the frame memory **1405**. In the encoding and decoding, the encoding/decoding engine **1407** uses the frame memory **1405** as a work area. Further, the encoding/decoding engine **1407** outputs a synchronization signal to the memory control section **1406** at a timing of start of processing on a macroblock basis, for example.

(398) The video ES buffer **1408A** buffers the video stream generated by the encoding/decoding engine **1407** and supplies the video stream to the multiplexing section (MUX) **1412**. The video ES buffer **1408B** buffers the video stream supplied from the demultiplexing section (DMUX) **1413** and supplies the video stream to the encoding/decoding engine **1407**.

(399) The audio ES buffer **1409A** buffers an audio stream generated by the audio encoder **1410** and supplies the audio stream to the multiplexing section (MUX) **1412**. The audio ES buffer **1409B** buffers the audio stream supplied from the demultiplexing section (DMUX) **1413** and supplies the audio stream to the audio decoder **1411**.

(400) The audio encoder **1410** performs, e.g., digital conversion on an audio signal input from, e.g., the connectivity **1321** (FIG. 37) and encodes the resultant signal by a predetermined system such as an MPEG audio system and an AC3 (AudioCode number 3) system. The audio encoder **1410** sequentially writes an audio stream, which is data in which the audio signal is encoded, in the audio ES buffer **1409A**. The audio decoder **1411** decodes the audio stream supplied from the audio ES buffer **1409B**, performs conversion into an analog signal, for example, and supplies the analog

signal, as a reproduced audio signal, to the connectivity **1321** (FIG. 37), for example.

(401) The multiplexing section (MUX) **1412** multiplexes the video stream and the audio stream. The multiplexing method (i.e., the format of a bit stream generated by multiplexing) is arbitrary. Further, at the time of multiplexing, the multiplexing section (MUX) **1412** can also add predetermined header information or the like to the bit stream. In other words, the multiplexing section (MUX) **1412** can convert the format of the stream by the multiplexing. For example, the multiplexing section (MUX) **1412** multiplexes the video stream and the audio stream to convert those streams into a transport stream, which is a bit stream having a format for transfer. Further, for example, the multiplexing section (MUX) **1412** multiplexes the video stream and the audio stream to convert those streams into data (file data) having a file format for recording.

(402) The demultiplexing section (DMUX) **1413** demultiplexes the bit stream, in which the video stream and the audio stream are multiplexed, by a method corresponding to the multiplexing by the multiplexing section (MUX) **1412**. In other words, the demultiplexing section (DMUX) **1413** extracts the video stream and the audio stream from the bit stream read from the stream buffer **1414** (separates the video stream and the audio stream from each other). In other words, the demultiplexing section (DMUX) **1413** can convert the format of the stream by the demultiplexing (inverse conversion of the conversion by the multiplexing section (MUX) **1412**). For example, the demultiplexing section (DMUX) **1413** can acquire the transport stream supplied from, e.g., the connectivity **1321** or the broadband modem **1333** (each of which is shown in FIG. 37) via the stream buffer **1414** and demultiplex the transport stream, to convert the transport stream into the video stream and the audio stream. Further, for example, the demultiplexing section (DMUX) **1413** can acquire the file data, which is read from various recording media by, e.g., the connectivity **1321** (FIG. 37), via the stream buffer **1414** and demultiplex the file data, to convert the file data into the video stream and the audio stream.

(403) The stream buffer **1414** buffers the bit stream. For example, the stream buffer **1414** buffers the transport stream supplied from the multiplexing section (MUX) **1412** and supplies the transport stream to, e.g., the connectivity **1321** or the broadband modem **1333** (each of which is shown in FIG. 37) at a predetermined timing or on the basis of a request from the outside or the like.

(404) Further, for example, the stream buffer **1414** buffers the file data supplied from the multiplexing section (MUX) **1412**, supplies the file data to, e.g., the connectivity **1321** (FIG. 37) at a predetermined timing or on the basis of a request from the outside or the like, and records the file data on various recording media.

(405) Moreover, the stream buffer **1414** buffers the transport stream acquired via, e.g., the connectivity **1321** or the broadband modem **1333** (each of which is shown in FIG. 37) and supplies the transport stream to the demultiplexing section (DMUX) **1413** at a predetermined timing or on the basis of a request from the outside or the like.

(406) Further, the stream buffer **1414** buffers the file data read from various recording media in, e.g., the connectivity **1321** (FIG. 37) and supplies the file data to the demultiplexing section (DMUX) **1413** at a predetermined timing or on the basis of a request from the outside or the like.

(407) Next, an operation example of the video processor **1332** having such a configuration will be described. For example, a video signal that is input to the video processor **1332** from the connectivity **1321** (FIG. 37) or the like is converted into digital image data of a predetermined system such as a 4:2:2Y/Cb/Cr system in the video input processing section **1401**, and then sequentially written in the frame memory **1405**. The digital image data is read by the first image zooming section **1402** or the second image zooming section **1403**, subjected to format conversion into a predetermined system such as a 4:2:0Y/Cb/Cr system and zooming processing, and written in the frame memory **1405** again. The image data is encoded by the encoding/decoding engine **1407** and written as a video stream in the video ES buffer **1408A**.

(408) Further, an audio signal that is input from the connectivity **1321** (FIG. 37) or the like to the video processor **1332** is encoded by the audio encoder **1410** and written, as an audio stream, in the

audio ES buffer **1409A**.

(409) The video stream of the video ES buffer **1408A** and the audio stream of the audio ES buffer **1409A** are read and multiplexed by the multiplexing section (MUX) **1412**, and converted into a transport stream, file data, or the like. The transport stream generated by the multiplexing section (MUX) **1412** is buffered by the stream buffer **1414**, and then output to an external network via, e.g., the connectivity **1321** or the broadband modem **1333** (each of which is shown in FIG. 37). Further, the file data generated by the multiplexing section (MUX) **1412** is buffered by the stream buffer **1414**, and then output to the connectivity **1321** (FIG. 37) or the like and recorded on various recording media.

(410) Further, the transport stream, which is input from the external network to the video processor **1332** via, e.g., the connectivity **1321** or the broadband modem **1333** (each of which is shown in FIG. 37), is buffered in the stream buffer **1414** and then demultiplexed by the demultiplexing section (DMUX) **1413**. Further, the file data, which is read from various recording media in, e.g., the connectivity **1321** (FIG. 37) and input to the video processor **1332**, is buffered in the stream buffer **1414** and then demultiplexed by the demultiplexing section (DMUX) **1413**. In other words, the transport stream or the file data input to the video processor **1332** is separated to be a video stream and an audio stream by the demultiplexing section (DMUX) **1413**.

(411) The audio stream is supplied to the audio decoder **1411** via the audio ES buffer **1409B** and decoded, to reproduce an audio signal. Further, the video stream is written in the video ES buffer **1408B**, and then sequentially read by the encoding/decoding engine **1407**, decoded, and written in the frame memory **1405**. The decoded image data is subjected to zooming processing by the second image zooming section **1403** and written in the frame memory **1405**. The decoded image data is then read by the video output processing section **1404**, subjected to format conversion of a predetermined system such as a 4:2:2Y/Cb/Cr system, and converted into an analog signal, to reproduce and output a video signal.

(412) In the case where the present disclosure is applied to the video processor **1332** configured as described above, the present disclosure according to each of the embodiments described above only needs to be applied to the encoding/decoding engine **1407**. In other words, for example, the encoding/decoding engine **1407** only needs to have a function of the encoding apparatus or the decoding apparatus according to the first to third embodiments. This enables the video processor **1332** to obtain an effect similar to that described above with reference to FIGS. 1 to 31.

(413) It should be noted that in the encoding/decoding engine **1407** the present disclosure (i.e., the function of the encoding apparatus or the decoding apparatus according to each of the embodiments described above) may be achieved by hardware such as a logic circuit, may be achieved by software such as an embedded program, or may be achieved by both of them.

(414) (Another Configuration Example of Video Processor)

(415) FIG. 39 shows another example of a schematic configuration of the video processor **1332** (FIG. 37) to which the present disclosure is applied. In the case of the example of FIG. 39, the video processor **1332** has a function of encoding/decoding video data by a predetermined system.

(416) More specifically, as shown in FIG. 39, the video processor **1332** includes a control section **1511**, a display interface **1512**, a display engine **1513**, an image processing engine **1514**, and an internal memory **1515**. Further, the video processor **1332** includes a codec engine **1516**, a memory interface **1517**, a multiplexing/demultiplexing section (MUX DMUX) **1518**, a network interface **1519**, and a video interface **1520**.

(417) The control section **1511** controls operations of respective processing sections within the video processor **1332**, such as the display interface **1512**, the display engine **1513**, the image processing engine **1514**, and the codec engine **1516**.

(418) As shown in FIG. 39, the control section **1511** includes, for example, a main CPU **1531**, a sub CPU **1532**, and a system controller **1533**. The main CPU **1531** executes a program for controlling the operations of the respective processing sections within the video processor **1332**, or the like.

The main CPU **1531** generates a control signal according to that program or the like and supplies the control signal to the processing sections (i.e., controls the operations of the respective processing sections). The sub CPU **1532** plays an auxiliary role for the main CPU **1531**. For example, the sub CPU **1532** executes a child process, a subroutine, or the like of the program executed by the main CPU **1531** or the like. The system controller **1533** controls operations of the main CPU **1531** and the sub CPU **1532**, such as specifying a program executed by the main CPU **1531** and the sub CPU **1532**.

(419) The display interface **1512** outputs image data to, e.g., the connectivity **1321** (FIG. 37) or the like under the control of the control section **1511**. For example, the display interface **1512** converts the image data of digital data into an analog signal and outputs the analog signal as a reproduced video signal or the image data of digital data without change, to a monitoring apparatus of the connectivity **1321** (FIG. 37) or the like.

(420) The display engine **1513** performs, under the control of the control section **1511**, various types of conversion processing such as format conversion, size conversion, and color gamut conversion on the image data so as to be matched with hardware specifications of a monitoring apparatus or the like that displays an image of the image data.

(421) The image processing engine **1514** performs predetermined image processing such as filter processing for, for example, image quality improvement, on the image data under the control of the control section **1511**.

(422) The internal memory **1515** is a memory that is provided inside the video processor **1332** and shared with the display engine **1513**, the image processing engine **1514**, and the codec engine **1516**. The internal memory **1515** is used for, for example, giving and receiving data between the display engine **1513**, the image processing engine **1514**, and the codec engine **1516**. For example, the internal memory **1515** stores data supplied from the display engine **1513**, the image processing engine **1514**, or the codec engine **1516** and supplies the data to the display engine **1513**, the image processing engine **1514**, or the codec engine **1516** as needed (for example, in response to a request). The internal memory **1515** may be achieved by any storage device. In general, since the internal memory **1515** is frequently used for storing a small amount of data such as image data in units of block and parameters, the internal memory **1515** may be desirably achieved by a semiconductor memory that has a relatively (compared with, for example, the external memory **1312**) small capacity but has a high speed response, such as an SRAM (Static Random Access Memory).

(423) The codec engine **1516** performs processing on encoding or decoding of the image data. An encoding/decoding system corresponding to the codec engine **1516** is arbitrary, and the number thereof may be one or more than one. For example, the codec engine **1516** may have codec functions of a plurality of encoding/decoding systems and perform encoding of image data or decoding of encoded data by a function selected from the above functions.

(424) In the example shown in FIG. 39, the codec engine **1516** includes, as functional blocks of processing on the codec, for example, an MPEG-2 Video **1541**, an AVC/H. 264 **1542**, an HEVC/H.265 **1543**, an HEVC/H.265 (Scalable) **1544**, an HEVC/H. 265 (Multi-view) **1545**, and an MPEG-DASH **1551**.

(425) The MPEG-2 Video **1541** is a functional block for encoding and decoding the image data by an MPEG-2 system. The AVC/H.264 **1542** is a functional block for encoding and decoding the image data by an AVC system. The HEVC/H. 265 **1543** is a functional block for encoding and decoding the image data by an HEVC. The HEVC/H. 265 (Scalable) **1544** is a functional block for scalable encoding and scalable decoding of the image data by the HEVC. The HEVC/H. 265 (Multi-view) **1545** is a functional block for multi-view encoding and multi-view decoding of the image data by the HEVC.

(426) The MPEG-DASH **1551** is a functional block for sending and receiving the image data by an MPEG-DASH (MPEG-Dynamic Adaptive Streaming over HTTP) system. The MPEG-DASH is a

video streaming technique using an HTTP (HyperText Transfer Protocol), and one of the features thereof is to select and transmit an appropriate one piece of encoded data in units of segment from pieces of the encoded data previously prepared that are different in resolutions or the like. The MPEG-DASH **1551** performs generation of a stream according to standards, transmission and control of the stream, and the like and uses the above-mentioned MPEG-2 Video **1541** to the HEVC/H.265 (Multi-view) **1545** for encoding/decoding of the image data.

(427) The memory interface **1517** is an interface for the external memory **1312**. Data supplied from the image processing engine **1514** or the codec engine **1516** is supplied to the external memory **1312** via the memory interface **1517**. Further, data read from the external memory **1312** is supplied to the video processor **1332** (the image processing engine **1514** or the codec engine **1516**) via the memory interface **1517**.

(428) The multiplexing/demultiplexing section (MUX DMUX) **1518** multiplexes or demultiplexes various types of data on images, such as a bit stream of encoded data, image data, and a video signal. The multiplexing/demultiplexing method is arbitrary. For example, at the time of multiplexing, the multiplexing/demultiplexing section (MUX DMUX) **1518** can not only collect pieces of data into one but also add predetermined header information or the like to that data. Further, at the time of demultiplexing, the multiplexing/demultiplexing section (MUX DMUX) **1518** can not only divide on piece of data into pieces of data but also add predetermined header information or the like to the divided pieces of data. In other words, the multiplexing/demultiplexing section (MUX DMUX) **1518** can convert a data format by multiplexing/demultiplexing. For example, by multiplexing a bit stream, the multiplexing/demultiplexing section (MUX DMUX) **1518** can convert the bit stream into a transport stream, which is a bit stream having a format for transfer, or data (file data) having a file format for recording. As a matter of course, inverse conversion thereof can also be performed by demultiplexing.

(429) The network interface **1519** is an interface for, for example, the broadband modem **1333**, the connectivity **1321** (each of which is shown in FIG. 37), or the like. The video interface **1520** is an interface for, for example, the connectivity **1321**, the camera **1322** (each of which is shown in FIG. 37), or the like.

(430) Next, an operation example of the video processor **1332** as described above will be described. For example, when a transport stream is received from an external network via the connectivity **1321**, the broadband modem **1333** (each of which is shown in FIG. 37), or the like, the transport stream is supplied to the multiplexing/demultiplexing section (MUX DMUX) **1518** via the network interface **1519** and demultiplexed, and then decoded by the codec engine **1516**. Image data obtained by the decoding of the codec engine **1516** is, for example, subjected to predetermined image processing by the image processing engine **1514**, subjected to predetermined conversion by the display engine **1513**, and supplied to, e.g., the connectivity **1321** (FIG. 37) or the like via the display interface **1512**, so that an image thereof is displayed on a monitor. Further, for example, image data obtained by the decoding of the codec engine **1516** is encoded again by the codec engine **1516**, multiplexed and converted into file data by the multiplexing/demultiplexing section (MUX DMUX) **1518**, output to, e.g., the connectivity **1321** (FIG. 37) or the like via the video interface **1520**, and recorded on various recording media.

(431) Moreover, for example, file data of encoded data, in which image data is encoded and which is read from a recording medium not shown in the figure by the connectivity **1321** (FIG. 37) or the like, is supplied to the multiplexing/demultiplexing section (MUX DMUX) **1518** via the video interface **1520** and demultiplexed, and then decoded by the codec engine **1516**. Image data obtained by the decoding of the codec engine **1516** is subjected to predetermined image processing by the image processing engine **1514**, subjected to predetermined conversion by the display engine **1513**, and supplied to, e.g., the connectivity **1321** (FIG. 37) or the like via the display interface **1512**, so that an image thereof is displayed on a monitor. Further, for example, image data obtained

by the decoding of the codec engine **1516** is encoded again by the codec engine **1516**, multiplexed and converted into a transport stream by the multiplexing/demultiplexing section (MUX DMUX) **1518**, supplied to, e.g., the connectivity **1321**, the broadband modem **1333** (each of which is shown in FIG. **37**), or the like via the network interface **1519**, and transmitted to another apparatus not shown in the figure.

(432) It should be noted that giving and receiving of image data or other data between the processing sections of the video processor **1332** is performed by using, for example, the internal memory **1515** or the external memory **1312**. Further, the power management module **1313** controls power supply to the control section **1511**, for example.

(433) In the case where the present disclosure is applied to the video processor **1332** configured as described above, the present disclosure according to each of the embodiments described above only needs to be applied to the codec engine **1516**. In other words, for example, the codec engine **1516** only needs to include a functional block that achieves the encoding apparatus or the decoding apparatus according to the first to third embodiments. With the codec engine **1516** thus configured, the video processor **1332** can obtain an effect similar to that described above with reference to FIGS. **1** to **31**.

(434) It should be noted that in the codec engine **1516** the present disclosure (i.e., the function of the encoding apparatus or the decoding apparatus according to each of the embodiments described above) may be achieved by hardware such as a logic circuit, may be achieved by software such as an embedded program, or may be achieved by both of them.

(435) The two examples of the configuration of the video processor **1332** have been described above, but the configuration of the video processor **1332** is arbitrary and may be other than the two examples described above. Further, the video processor **1332** may be configured as one semiconductor chip, but may be configured as a plurality of semiconductor chips. For example, the video processor **1332** may be a three-dimensional stacked LSI that stacks a plurality of semiconductors. Further, the video processor **1332** may be achieved by a plurality of LSIs.

(436) (Example of Application to Apparatus)

(437) The video set **1300** can be incorporated into various apparatuses that process image data. For example, the video set **1300** can be incorporated into the television apparatus **900** (FIG. **33**), the mobile phone **920** (FIG. **34**), the recording/reproduction apparatus **940** (FIG. **35**), the imaging apparatus **960** (FIG. **36**), or the like. By incorporating the video set **1300** therein, the apparatus can obtain an effect similar to that described above with reference to FIGS. **1** to **31**.

(438) It should be noted that if a part of each configuration of the video set **1300** described above includes the video processor **1332**, the part can be carried out as a configuration to which the present disclosure is applied. For example, only the video processor **1332** can be carried out as a video processor to which the present disclosure is applied. Further, for example, as described above, the processor indicated by the dotted line **1341**, the video module **1311**, or the like can be carried out as a processor, a module, or the like to which the present disclosure is applied.

Moreover, for example, it is also possible to combine the video module **1311**, the external memory **1312**, the power management module **1313**, and the front end module **1314** to be carried out as a video unit **1361** to which the present disclosure is applied. In the case of any configuration, it is possible to obtain an effect similar to that described above with reference to FIGS. **1** to **31**.

(439) In other words, similarly to the case of the video set **1300**, any configuration including the video processor **1332** can be incorporated in various apparatuses that process image data. For example, the video processor **1332**, the processor indicated by the dotted line **1341**, the video module **1311**, or the video unit **1361** can be incorporated in the television apparatus **900** (FIG. **33**), the mobile phone **920** (FIG. **34**), the recording/reproduction apparatus **940** (FIG. **35**), the imaging apparatus **960** (FIG. **36**), or the like. By incorporating any configuration to which the present disclosure is applied, such an apparatus can obtain an effect similar to that described above with reference to FIGS. **1** to **31**, similarly to the case of the video set **1300**.

(440) It should be noted that the example in which various types of information are multiplexed into encoded data and transmitted from the encoding side to the decoding side has been described in this specification. However, the technique of transmitting those pieces of information is not limited to such an example. For example, those pieces of information may be transmitted or recorded as individual pieces of data associated with encoded data, without being multiplexed into the encoded data. Here, the term “associated” means that an image included in a bit stream (which may be a part of an image, such as slice and block) and information corresponding to that image can be linked to each other at the time of decoding. In other words, the information may be transmitted through a transmission path different from that for the encoded data. Further, the information may be recorded on a recording medium (or another recording area of the same recording medium) different from that for the encoded data. Moreover, the information and the encoded data may be associated with each other in arbitrary units of, e.g., a plurality of frames, one frame, or a part of a frame.

(441) The present disclosure can be applied to an encoding apparatus or a decoding apparatus that is used when, as in MPEG, H.26x, or the like, a bit stream compressed by orthogonal transformation such as discrete cosine transform and motion compensation is received via a network medium such as satellite broadcasting, a cable TV, the Internet, and a mobile phone or is processed on a storage medium such as an optical disc, a magnetic disk, and a flash memory.

(442) Further, in this specification, the system means an aggregation of a plurality of constituent elements (apparatus, module (component), and the like), regardless of whether all constituent elements are included in the same casing. Therefore, a plurality of apparatuses accommodated in separate casings and connected to one another via a network is a system, and one apparatus including a plurality of modules in one casing is also a system.

(443) Moreover, the effects described in this specification are merely exemplary ones and are not restrictive ones, and any other effects may be produced.

(444) Further, the embodiments of the present disclosure are not limited to the embodiments described above and can be variously modified without departing from the gist of the present disclosure.

(445) For example, in the first and second embodiments, the position of the picture identifying information of the current picture registered in the reference picture list may not be fixed. Further, in the first and third embodiments, the picture type of the current picture set in the reference picture list may be STRP.

(446) Further, in the second embodiment, as in the third embodiment, the picture identifying information of the current picture may be registered not in the reference picture list but in the temporal list.

(447) Moreover, in the third embodiment, as in the second embodiment, by a change of the method of generating a predicted vector list in the case where the type of the reference picture of the current picture is Intra BC, coding efficiency in the case where Intra BC and inter coding are communalized may be improved.

(448) Further, for example, the present disclosure can have a configuration of cloud computing in which a plurality of apparatuses share one function and cooperate to perform processing via a network.

(449) Further, the steps described in the flowcharts described above can be executed by one apparatus or shared and executed by a plurality of apparatuses.

(450) Moreover, in the case where one step includes a plurality of processing steps, the plurality of processing steps in one step can be executed by one apparatus or shared and executed by a plurality of apparatuses.

(451) The present disclosure can also have the following configurations. (1) An image processing apparatus, including: a predicted vector generation section that sets, at a time of encoding of a current motion vector of a current block for a prediction using a correlation within a screen, in a

case where a type of a reference picture of the current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other, the candidate block to not available and generates a predicted vector of the current motion vector by using the reference motion vector, the reference motion vector being referred to when the predicted vector of the current motion vector is generated; and a difference vector generation section that generates a difference vector between the current motion vector and the predicted vector generated by the predicted vector generation section. (2) The image processing apparatus according to (1), further including a setting section that sets the type of the reference picture of the current block to a long-term reference picture. (3) The image processing apparatus according to (1), in which the predicted vector generation section determines that the type of the reference picture of the current block and the type of the reference picture of the candidate block are different from each other in a case where a picture of the candidate block and the reference picture of the candidate block are different from each other. (4) The image processing apparatus according to any one of (1) to (3), further including a list creation section that registers a picture of the current block in a predetermined position of a list of candidates of the reference picture of the current block. (5) The image processing apparatus according to (4), in which the predetermined position is a head. (6) The image processing apparatus according to (4) or (5), in which the list creation section rearranges order of the candidates of the reference picture in the list, before registering the picture of the current block. (7) The image processing apparatus according to (4) or (5), in which the list creation section rearranges order of the candidates of the reference picture other than the picture of the current block in the list, after registering the picture of the current block. (8) An image processing method for an image processing apparatus, the image processing method including: a predicted vector generation step of setting, at a time of encoding of a current motion vector of a current block for a prediction using a correlation within a screen, in a case where a type of a reference picture of the current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other, the candidate block to not available and generates a predicted vector of the current motion vector by using the reference motion vector, the reference motion vector being referred to when the predicted vector of the current motion vector is generated; and a difference vector generation step of generating a difference vector between the current motion vector and the predicted vector generated by processing of the predicted vector generation step. (9) An image processing apparatus, including: a predicted vector generation section that sets, at a time of decoding of a current motion vector of a current block for a prediction using a correlation within a screen, in a case where a type of a reference picture of the current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other, the candidate block to not available and generates a predicted vector of the current motion vector by using the reference motion vector, the reference motion vector being referred to when the predicted vector of the current motion vector is generated; and a motion vector generation section that adds a difference vector between the current motion vector and the predicted vector to the predicted vector generated by the predicted vector generation section and generates the current motion vector. (10) The image processing apparatus according to (9), further including a setting section that sets the type of the reference picture of the current block to a long-term reference picture. (11) The image processing apparatus according to (9), in which the predicted vector generation section determines that the type of the reference picture of the current block and the type of the reference picture of the candidate block are different from each other in a case where a picture of the candidate block and the reference picture of the candidate block are different from each other. (12) The image processing apparatus according to any one of (9) to (11), further including a list creation section that registers a picture of the current block in a predetermined position of a list of candidates of the reference picture of the current block. (13) The image processing apparatus according to (12), in which the predetermined position is a head. (14) The image processing apparatus according to (12)

or (13), in which the list creation section rearranges order of the candidates of the reference picture in the list, before registering the picture of the current block. (15) The image processing apparatus according to (12) or (13), in which the list creation section rearranges order of the candidates of the reference picture other than the picture of the current block in the list, after registering the picture of the current block. (16) An image processing method for an image processing apparatus, the image processing method including: a predicted vector generation step of setting, at a time of decoding of a current motion vector of a current block for a prediction using a correlation within a screen, in a case where a type of a reference picture of the current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other, the candidate block to not available and generates a predicted vector of the current motion vector by using the reference motion vector, the reference motion vector being referred to when the predicted vector of the current motion vector is generated; and a motion vector generation step of adding a difference vector between the current motion vector and the predicted vector to the predicted vector generated by processing of the predicted vector generation step and generating the current motion vector.

REFERENCE SIGNS LIST

(452) **50** encoding apparatus **88** list creation section **90** predicted vector generation section **91** difference vector generation section **110** decoding apparatus **144** predicted vector generation section **145** motion vector generation section **146** list creation section

Claims

1. An image processing apparatus, comprising: circuitry configured to exclude, in a case where a type of a reference picture of a current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other and neither the type of the reference picture of the current block nor the type of the reference picture of the candidate block is Short-Term Reference Picture (STRP), a motion vector of the candidate block from candidates of the reference motion vector, generate, in a case where a type of the reference picture of the current block and the type of the reference picture of the candidate block are each STRP, a predicted vector of a current motion vector by performing scaling on the reference motion vector on a basis of a difference in picture order count (POC) between the reference picture of the current block and the reference picture of the candidate block, generate, in a case where the type of the reference picture of the current block and the type of the reference picture of the candidate block are each Intra Block Copy (Intra BC), the predicted vector of the current motion vector by setting the reference motion vector to the predicted vector without scaling, generate, in a case where the type of the reference picture of the current block and the type of the reference picture of the candidate block are each Long-Term Reference Picture (LTRP), the predicted vector of the current motion vector by setting the reference motion vector to the predicted vector without scaling, register a picture of the current block in a predetermined position of a list of candidates of the reference picture of the current block, and add a difference vector between the current motion vector and the predicted vector to the predicted vector and generate the current motion vector.
2. The image processing apparatus according to claim 1, wherein the circuitry is further configured to determine that the type of the reference picture of the current block and the type of the reference picture of the candidate block are different from each other in a case where a picture of the candidate block and the reference picture of the candidate block are different from each other.
3. The image processing apparatus according to claim 2, wherein the predetermined position is a head of the list of candidates.
4. The image processing apparatus according to claim 1, wherein the predetermined position is a head of the list of candidates.
5. The image processing apparatus according to claim 1, wherein the circuitry is further configured

to rearrange order of the candidates of the reference picture in the list, before registering the picture of the current block.

6. The image processing apparatus according to claim 2, wherein the circuitry is further configured to rearrange order of the candidates of the reference picture in the list, before registering the picture of the current block.

7. The image processing apparatus according to claim 3, wherein the circuitry is further configured to rearrange order of the candidates of the reference picture in the list, before registering the picture of the current block.

8. The image processing apparatus according to claim 4, wherein the circuitry is further configured to rearrange order of the candidates of the reference picture in the list, before registering the picture of the current block.

9. The image processing apparatus according to claim 1, wherein the circuitry is further configured to rearrange order of the candidates of the reference picture other than the picture of the current block in the list, after registering the picture of the current block.

10. The image processing apparatus according to claim 2, wherein the circuitry is further configured to rearrange order of the candidates of the reference picture other than the picture of the current block in the list, after registering the picture of the current block.

11. The image processing apparatus according to claim 3, wherein the circuitry is further configured to rearrange order of the candidates of the reference picture other than the picture of the current block in the list, after registering the picture of the current block.

12. An image processing method comprising: excluding, in a case where a type of a reference picture of a current block and a type of a reference picture of a candidate block corresponding to a candidate of a reference motion vector are different from each other and neither the type of the reference picture of the current block nor the type of the reference picture of the candidate block is Short-Term Reference Picture (STRP), a motion vector of the candidate block from candidates of the reference motion vector; generating, in a case where a type of the reference picture of the current block and the type of the reference picture of the candidate block are each STRP, a predicted vector of a current motion vector by performing scaling on the reference motion vector on a basis of a difference in picture order count (POC) between the reference picture of the current block and the reference picture of the candidate block; generating, in a case where the type of the reference picture of the current block and the type of the reference picture of the candidate block are each Intra Block Copy (Intra BC), the predicted vector of the current motion vector by setting the reference motion vector to the predicted vector without scaling; generating, in a case where the type of the reference picture of the current block and the type of the reference picture of the candidate block are each Long-Term Reference Picture (LTRP), the predicted vector of the current motion vector by setting the reference motion vector to the predicted vector without scaling; registering a picture of the current block in a predetermined position of a list of candidates of the reference picture of the current block; adding a difference vector between the current motion vector and the predicted vector to the predicted vector; and generating the current motion vector using the predicted vector and the list.
